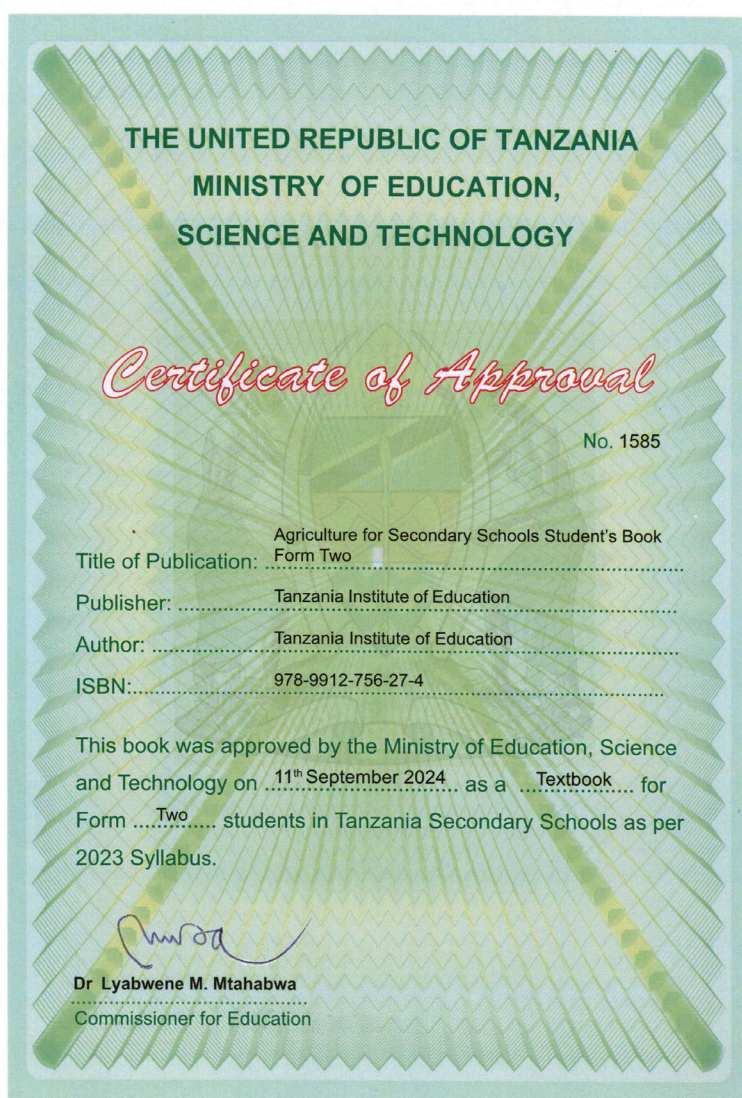


Agriculture

for Secondary Schools

Student's Book

Form Two



Tanzania Institute of Education

© Tanzania Institute of Education 2024

Published 2024

ISBN: 978-9912-756-27-4

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Abbreviations and Acronyms

a.s.l	above the sea level
AI	Artificial Insemination
BBTV	Banana Bunchy Top Virus
CAN	Calcium Ammonium Nitrate
CBSD	Cassava Brown Streak Disease
CMD	Cassava Mosaic Disease
DAP	Diammonium Phosphate
FYM	Farmyard Manure
GAPs	Good Agronomic Practices
IHT	Ideal Harvesting Time
IPM	Integrated Pest Management
JICA	Japan International Cooperation Agency
LITA	Livestock Training Agency
MATI	Ministry of Agriculture Training Institutes
SUA	Sokoine University of Agriculture
TARI	Tanzania Agricultural Research Institute
tmr	total mixed ration
TOSCI	Tanzania Official Seed Certification Institute
TSP	Triple Super Phosphate

Acknowledgements

The Tanzania Institute of Education (TIE) would like to acknowledge the contributions of all the organisations and individuals who participated in designing and developing this textbook. In particular, TIE wishes to thank the Sokoine University of Agriculture (SUA), Livestock Training Agencies (LITA), Ministry of Agriculture Training Institutes (MATI), the School Quality Assurance (SQA) Department, Teachers' Colleges and Secondary Schools. Besides, the following individuals are acknowledged:

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TIE also appreciates the contributions of the secondary school teachers and students who participated in the manuscript's trial phase. Likewise, the Institute would like to thank the Government of the United Republic of Tanzania for financing the writing and printing of this textbook.



Dr Aneth A. Komba
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Preface

This textbook, *Agriculture for Secondary Schools*, is explicitly written for Form Two students in the United Republic of Tanzania. It is prepared according to the 2023 Agriculture Syllabus for Ordinary Secondary Education Form I - IV issued by the Ministry of Education, Science and Technology. Some of the contents in this book have been adapted from the Agriculture for Secondary Schools Student's Book Form One (2021), Form Two (2021), and Form Three (2022) which were published in accordance with the 2019 agriculture syllabus for Ordinary Level Secondary Education, issued by the Ministry of Education Science and Technology.

The book consists of eight chapters namely, General principles and practices of the production of crops, Banana production, Cassava production, Sweet potato production, and Round potato production. Other chapters are General principles and practices of livestock production, Pig production, and Rabbit production. Each chapter contains texts, illustrations, activities, and exercises. You are encouraged to do all the activities and attempt all the exercises and other assignments provided by your teacher. Doing so will enable you to develop the intended competencies.

Additional learning resources are available in the TIE e-Library at <https://ol.tie.go.tz> or ol.tie.go.tz



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Chapter One

General principles and practices of crops production

Introduction

This chapter will introduce you to the principles and practices applied in the production of crops. The competencies developed from this chapter will enable you to use the principles and practices when producing various crops for commercial and domestic purposes.



Think

“Crop farming without adhering to principles and GAPs, is like navigating unknown sea without a map”.

Importance of principles and practices in crop production

Good agricultural practices in crop production are crucial for farmers and other people engaged in agricultural-related fields, such as those working in gardening, plant science, research, food processors, distributors, and sellers. Understanding the principles and practices helps an individual produce crops efficiently and sustainably and, therefore, a successful crop farming business.

General principles and practices of crop production

Principles and practices of crop production are the scientific theories, ways, methods, and rules of producing crops. It involves basic factors, theories, cultural operations, and practices that govern profitable crop production in terms of growth, development, and yield or revenue accrued after input costs have been removed from total revenue. To produce any crop profitably and sustainably, it is recommended to adhere to the principles and practices of crop production.

The principles and practices of crop production involve:

- (a) Planning for crop production;
- (b) Land preparation;
- (c) Selection of planting materials and sowing/planting;
- (d) Management of soil fertility;
- (e) Management of soil water;
- (f) Management of pests;
- (g) Performing other specific agronomic practices;
- (h) Harvesting and post-harvesting management; and
- (i) Farm record keeping.

Planning for crop production

Planning is the foremost mental activity in crop production. This stage gives one an insight into the whole picture of the intended agricultural business from the beginning to the end of a farming enterprise. It allows one to determine the physical and financial resources required for the enterprise. A planning stage enables an individual to determine and focus on the reasons behind engaging in the production of the intended crop, find out the market for the crop before starting to produce it and determine the required production resources such as land availability, labour, and financial resources.

In recent years, planning for agricultural production has been threatened by climate change experienced in the form of increasing temperatures, longer dry spells, prolonged drought, floods, and erratic rainfall. Therefore, planning in the face of climate change needs to consider minimising the risk of climate change impacts: This can be done by including several climate-smart agricultural practices, such as.

- (a) Diversifying crops and livestock: Involving drought-tolerant crops in the farm can be one of the most effective ways for a farmer to adapt to climate change. It can help prevent crop failures, maintain soil health, and prevent the spread of pests and diseases.
- (b) Using local seed varieties: Small landholder farmers can use crop cultivars that can withstand local pest pressure, water and nutrient deficiency.
- (c) Implementing sustainable agricultural practices: These include conservation agriculture and agroforestry practices that can help increase resistance to climate change by improving soil health, water retention, and biodiversity.

- (d) Prioritising climate-smart strategies and investments: This includes financial services like credit and insurance, which can help farmers manage climate-related risks, such as managing drought through irrigation.
- (e) Improving food storage and distribution: This involves using climate-smart storage and transportation facilities to avoid crop deterioration due to bad weather.
- (f) Breeding for climate adaptive crops: This involves developing crop varieties resistant to climate-related hazards (drought, floods, pests) to help farmers adapt to climate change.

Activity 1.1

1. Design a project focused on any crop. Develop a detailed crop production plan covering planning, site selection, land preparation, selection of planting materials, planting, water, nutrients and pest management, harvesting, and post-harvesting handling.
2. Inquire from elders and experienced farmers on historical trends of rainfall distribution and its influence on crop production in your local area.
3. Summarise what you have learned from this Activity in your portfolio.

Site selection

Proper field site selection is one of the most effective ways to reduce crop production risks. It involves evaluating factors that support optimal crop growth and productivity. Key considerations include climate, weather, soil, water availability, topography, pest and disease pressure, accessibility, and infrastructure. By carefully assessing these aspects, the best location can be chosen to maximise crop growth and improve outcomes.

Activity 1.2

1. Visit your school farm and assess the suitability of the following factors for producing a crop of your choice.
 - (a) Soil;
 - (b) water availability/rainfall;
 - (c) temperature;
 - (d) topography; and
 - (e) pest pressure.

2. Using the factors above, suggest suitable areas for cassava, round potato, and banana production.

Land preparation

Land preparation involves all the activities that make the land suitable for planting and facilitate the germination or growth of planting materials and their development. For rain-fed agriculture, land should be prepared well before the onset of rains to give weeds and other vegetation enough time to dry up and decompose into organic matter. Also, it needs enough time to allow carbon dioxide and other gases to diffuse out of the soil while being replaced with oxygen, which is essential for seed germination and the growth of beneficial soil organisms. Early land preparation gives enough time for other subsequent operations to be done, giving way to early planting. The proper time for land preparation differs with the crop type; virgin land usually needs longer than non-virgin land. This is because sufficient time is required for land clearing, ploughing, and harrowing.

Tillage

Land must often be subjected to some form of tillage before planting the crop. Tillage involves changing soil conditions with a tool/implement to obtain ideal conditions for seed germination, seedling establishment, and crop growth.

Tillage operations, done before planting, are collectively referred to as preparatory tillage. They involve deep opening, loosening, or other soil manipulation to bring about a desirable tilth, depending on the nature of the land and crop to be planted. The tilth may be coarse, moderate, or fine, depending on the crop's requirement for growth.

Activity 1.3

1. Visit your school farm/nearby crop fields in your area, and or use internet/video clips to study land preparatory activities done before planting a crop.
2. Then, use the following table to fill in, the observed land preparatory activities that take place before planting the studied crop.

Name of the crop you have studied.....

Location/place where the crop is grown.....

The proper time for preparing land before planting the crop.....

S/ No.	Tillage operational practice	Tool/ implement used	Reason for carrying out the practice

- Summarise the lesson learned from this activity in your portfolio.
- Adopt and apply the learned land preparatory practices in the school farm.

Exercise 1.1

- Use various learning resources such as a library and internet searches to find out the meaning of the following terms as applied in land preparation for growing crops:
 - ploughing;
 - harrowing;
 - ridging;
 - mounding ;
 - terracing; and
 - bed making.
- Outline various tools and implements that are used in land clearing and preparation.
- Find at least five reasons why tillage is an important land preparatory practice before planting a crop.

Selection of planting materials

Selecting the proper planting seeds or materials is crucial for successful crop production. While some crops, like maize, rely on seeds, others use different planting materials such as cuttings for cassava, vines for sweet potatoes, and suckers for banana and sisal. The planting materials should be clean and sourced from recommended institutions such as the Agricultural Seed Agency (ASA), Tanzania Agricultural Research Institute (TARI), and private seed companies, verified by the Tanzania Official Seed Certification Institute (TOSCI). Key factors in determining the quality of seeds or planting materials include germination

percentage, adaptability to local conditions, resistance to pests and diseases, yield potential, and maturity period. Farmers can increase crop performance and, eventually, high yield by focusing on these factors.

Planting crops

Planting a crop is an operation that follows after land preparation. The practice involves placing a planting material in the soil or other planting media to initiate the growth or establishment of a crop. Planting is sometimes referred to as propagation. However, the process is called sowing when seeds are used as planting material. Planting crops using seeds is also termed generative or sexual propagation while establishing crops using vegetative materials is called asexual or vegetative propagation. Crops established by sowing seeds include maize, groundnuts, sorghum, millets, wheat, cowpeas, and watermelon. Crops such as yams, sugar cane, ginger, sisal, cocoyam/taro, sweet potatoes, and cassava are typically established by planting vegetative parts. Therefore, the farmer often has to opt for using seeds or vegetative planting materials, depending on the crop and the simplicity of the crop, to germinate (for seeds) or sprout (for other vegetative materials). Figure 1.1 (a)- (f) represents some of the planting materials; study them, then carry out an Activity 1.4.



Figure 1.1 (a)



Figure 1.1 (b)



Figure 1.1 (c)



Figure 1.1 (d)

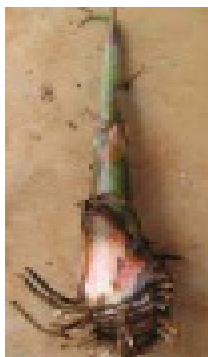


Figure 1.1 (e)



Figure 1.1 (f)

Figure 1.1: Planting materials

Activity 1.4

- Referring to the pictures represented in Figure 1.1, perform the following tasks:
 - Identify each of the planting materials presented in Figure 1.1 (a)-(f) above;
 - Categorise the planting materials presented in Figure 1.1 into:
 - Vegetative planting materials; and
 - Sexual planting materials.
- Using a table, compare the advantages and disadvantages of vegetative against sexual planting materials.

After preparing a seedbed and planting materials, there are factors that must be considered before planting. These include spacing between and within stands, plant population and seed rate, number of seeds per stand, depth to which seeds are placed, and seed position with respect to the tillage operations.

Plant population and seed rate determination

Plant population is the number of plants per unit area. Sometimes, it is termed as plant density. An optimum plant population gives plants enough resources for growth and development. Such resources include soil nutrients, moisture, light, and air. This leads to high crop yields if other crop production factors are kept constant. Plant overpopulation and under population often result in reduced yields. Overpopulation of crops results in competition for resources, whereas in under-population, the environmental resources are under-utilised. Plant population is

determined by dividing the planting area by the spacing of the crop.

$$\text{Plant population} = \frac{\text{Area of the land}}{\text{Spacing of the crop}}$$

The area of one hectare of land is equals to 10,000 square metres. The time of planting a given crop is essential as it influences the quantity and quality of the crop produced. Time factor influences rainfall, temperature, occurrence of pests, marketing of the crop, and cropping system.

Methods of planting crops

Crops are generally planted in a field using several methods, including broadcasting, row planting, drilling, transplanting, over-sowing, and under-sowing.

Activity 1.5

1. Visit a school farm/nearby crop fields, and nurseries; and or use the internet and relevant video clips to inquire and study on:
 - (a) Different planting materials, including grafted and budded planting materials;
 - (b) Various methods used in planting crops; and
 - (c) Tools and implements used in planting the crops.
2. Pay a study visit on cassava /sugarcane/sweet potatoes and learn how cuttings/setts/vines/suckers are prepared.
3. Visit some nearby seed dealers and inquire about how seeds are obtained and the criteria for good-quality seeds.
4. Identify entrepreneurial opportunities available in multiplying/developing planting materials.

Exercise 1.2

1. A farmer intends to plant maize on one acre of land. If the spacing for the maize is 75 cm between rows and 25 cm between plants, determine the plant population for this field.
2. What is the relationship between plant population and crop yield?
3. How does sowing depth influence seed germination?
4. Advise a farmer:
 - (a) what to do for an overpopulated sunflower crop field, just two weeks after sowing.
 - (b) on effects of overpopulated crop plants in the field.

Nutrient management

Like other living things, plants need nutrients for proper growth and produce high yields. Plant nutrients are chemical elements that are essential to the nourishment of plant health. Soil is among the most crucial land resource components in crop production. Apart from providing anchorage to the plants, air, and water, it also provides nutrients for the proper growth and development of the crops. The soil that can supply plants with adequate nutrients is fertile. Therefore, soil fertility is the ability of the soil to provide crop plants with the required nutrients in proper proportions for optimal production. A decrease or deficiency in the nutrients crop plants need, affects growth and leads to low yields.

About sixteen nutrients are essential for normal plant growth. These nutrients must be supplied to plants in sufficient quantities and proportions. Among these, some nutrients are required by plants in large quantities, while others are needed in small amounts. Those required in large quantities are termed macro-nutrients, while those needed in small amounts are called micro-nutrients or trace-nutrients. Plant nutrients are also called elements. Figure 1.2 shows the classification of plant nutrients.

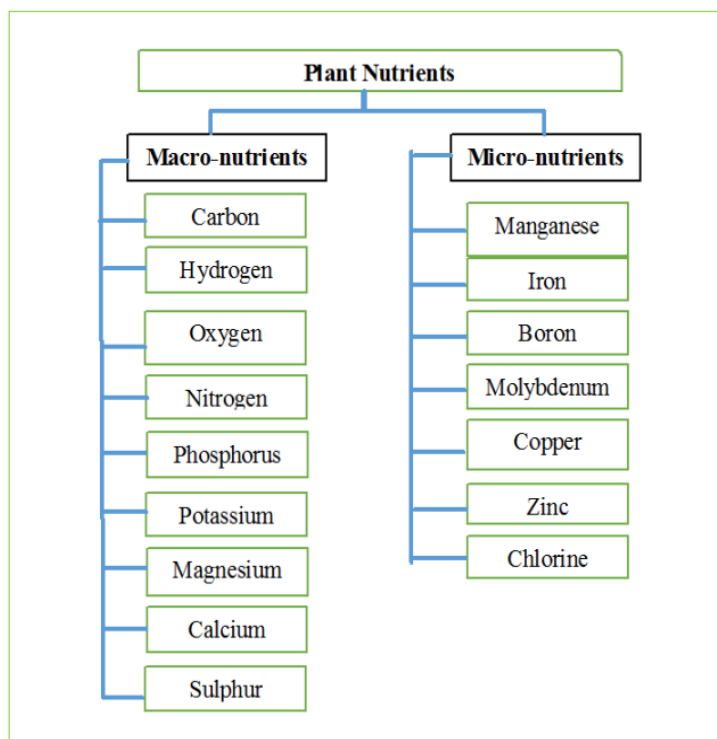


Figure 1.2: Classification of plant nutrients

Activity 1.6

1. Visit a school farm or nearby crop fields, then perform the following tasks:
 - (a) Observe different nutrients management practices employed in the field;
 - (b) Participate in carrying out some of the management practices employed to the crop plants;
 - (c) Adopt any two best practices, and apply them to crop plants in the school farm.
2. Summarise the lesson learned from this activity, in your portfolio.

Loss of soil fertility

Nutrients in the soil can be depleted, making the soil infertile enough to affect crop productivity. There are various means by which soil can lose its fertility. The most common causes of loss of soil fertility include soil erosion, burning of vegetation cover, water logging, leaching, flooding, and crop removal through continuous cropping and grazing. Others include changes in soil pH due to poor agronomic practices, accumulation of salts, volatilisation, and use by soil microorganisms.

Methods of maintaining soil fertility

There are various practices by which soil fertility can be retained in the soil. These include:

- (a) The use of good agronomic practices (GAPs) such as crop rotation, mulching, contour farming, proper irrigation, intercropping, agroforestry, reforestation, and use of cover crops and windbreaks;
- (b) Use of materials containing organic matters; and
- (c) Use of materials containing plant-concentrated nutrients in the soil.

Activity 1.7

1. Visit a school farm or some nearby crop fields and or use a library/online sources /video clips to study the following:
 - (a) Different ways through which soil can lose its fertility; and
 - (b) Various methods used to restore/ improve soil fertility.

2. Use your school farm/home set-up:
 - (a) To practice some good agronomic practices on maintaining soil fertility; and
 - (b) In a tabular form, show how each agronomic practice you carried out contributes to the maintenance of soil fertility.
3. Summarise what you have learnt from this activity in your portfolio.

Soil water management

Soil water or soil moisture refers to the soil held between spaces of soil particles. Usually, soil water contains dissolved nutrients to form a soil solution, which is a source of water and nutrients for plants. Water is essential for the growth and development of crop plants. Too much or too little water in the soil influences crop plant development and productivity.

Plants lose water in the form of water vapour from their surfaces. This process is termed transpiration. It occurs mainly at the surface of plants, particularly from leaves. If this water is not replaced, plants will wilt and eventually die. Water is also lost from the soil surface, especially during daytime, by evaporation. As water evaporates, the amount of water in the soil is reduced. As a result, plant roots find it more difficult to absorb water from the soil. If water is not added to the soil through irrigation or rain, plants will wilt and die. Water must, therefore, be present in the soil at an optimal level to prevent plants from wilting.

Critical stages of crops' moisture requirements

Crop plants have growth and development stages, which are more sensitive to soil moisture stress. The term critical period is commonly used to define the growth stage when plants are most sensitive to a water shortage. If a farmer needs to realise high yields, these stages should be known for effective production. This is because inadequate water supply during moisture-sensitive periods will irreversibly reduce the yield. Hence, the provision of adequate water to crop plants during such periods or stages is essential to avoid yield loss that can occur. For most crops, these stages are germination or establishment and grain/tuber or seed formation.

Conservation of soil moisture

Conservation of soil moisture is essential for maintaining soil water for agricultural production. It is mainly targeted to minimise the water lost from soils through evapotranspiration, percolation, and run-off. Run-off occurs when the rainfall intensity exceeds the soil's infiltration capacity, which measures the soil's ability to absorb and transmit rainwater down the soil profile. Conservation of soil moisture also helps to minimise the irrigation needs of crops. Various methods can be used to conserve soil moisture. Most methods rely on providing covers for the soil or other means to minimise water loss through evaporation, percolation, and run-off. It is worth noting that most methods used for conserving and improving soil fertility also apply here. These include:

- (a) application of well-decomposed compost, farmyard manure, or crop residue over the soil;
- (b) placing mulches on the soil;
- (c) crop rotation;
- (d) practising mixed cropping and inter-planting; and
- (e) terracing.

Management of deficit of soil moisture levels

When soil moisture levels are low, water must be applied to the soil to supply sufficient moisture essential for crop growth. This is referred to as irrigation. There are four main types of irrigation: surface, subsurface, drip, and overhead.

Management of excessive water levels in the soil

During rain or irrigation, the fields become wet. Rain/irrigation water penetrates the soil and is stored in its spaces. When all the pore spaces are filled with water, the soil is said to be saturated, and no more water can be added. Except for some crops, such as paddy and taro, which withstand saturated soil for long periods, a prolonged saturation state in the soil compromises the plant's health. To address the challenge of excess water in the field, the crop fields must have drainage systems installed, making those soils even more productive. Drainage of wet fields overcomes water-logged conditions and allows for a longer growing season. Several mechanisms such as dikes and soil bunds can help protect fields from unexpected floods and enable farmers to drain or retain water as needed; drainage ditches can channel away excess floodwaters.

Activity 1.8

Re-visit your school farm and nearby crop fields and or use ICT devices to perform the following tasks:

1. Observe and identify different ways by which:
 - (a) soil moisture is conserved;
 - (b) deficit of soil moisture is managed, and
 - (c) excessive water is managed in the field.
2. Based on your field experiences, library, ICT, and internet searches, outline various technologies for managing water in the crop field.
3. Summarise your observations in your portfolio.

Exercise 1.3

1. Soil water is vital to crop plants in several ways. Using various sources of information, outline any five roles/functions of soil water to a crop plant.
2. Identify the critical stages during which the crop demands significant water.
3. What crops require minimal water and can be grown in areas with limited water availability?
4. How does planting cover crops and mulch help to retain moisture in the soil?
5. Recommend to farmers any two methods for managing water on a steeply sloped crop field.

Pest management in crop production

Crop pest refers to a living organism that harms crops directly or indirectly. Crop pests include insects, rodents, mites, weeds, birds, and large animals (vermin), including monkeys, elephants, and wild pigs. Plant pathogens (e.g., bacteria, viruses, fungi, and nematodes) cause plant diseases such as bacterial wilt, damping off, Cassava Mosaic disease, and stunting. Pests can directly impact plants by feeding on them, which can cause damage to leaves, stems, and other parts of the plant. Pests can also indirectly impact plants by changing the host plant in ways that affects interaction with other pests. For example, pathogens can affect plants' growth and fitness and induce plant changes that impact susceptibility to insect pests. Insects can indirectly impact plants by transmitting pathogens as they feed on host plants. Large animals (e.g., monkeys, elephants, wild pigs, rodents, quelea birds are pests that destroy crops by feeding on them. Figure 1.3 (a)-(i) presents crop pests in different environments. Study them, and carry out Activity 1.7.



Figure 1.3 (a)



Figure 1.3 (b)



Figure 1.3 (c)



Figure 1.3 (d)



Figure 1.3 (e)



Figure 1.3 (f)



Figure 1.3 (g)



Figure 1.3 (h)



Figure 1.3 (i)

Figure 1.3: Some crop pests found on fields

Activity 1.9

1. Refer to Figure 1.3 (a) – (i) perform the following tasks:
 - (a) Identify each crop pest in the Figure;
 - (b) Explain why the above organisms in Figure 1.3 are considered pests;
 - (c) Identify crops and the stage of crop production at which the above pests harm the crop; and
 - (d) Suggest how each of the above pests can be controlled/managed.
2. Visit your school/home farm, food stores, and some nearby crop fields; observe and identify:
 - (a) pests and their effect on crops when they are in the field; and
 - (b) pests and their impact on crops when the produce is in storage.
3. Summarise your work in your portfolio.

Harvesting crops

Matured crops must be harvested from the field and processed for immediate use, storage or marketing. Harvesting is the process of collecting/gathering the entire plants or economic plant parts for economic purposes after attaining physiological maturity. Harvesting can be for direct eating, processing, storage, or marketing. Harvesting has to be done at the right time to ensure that the potential maximum yield and quality of the crops is fully realised as soon as possible after the physiological maturity stage has been reached. The physiological maturity of a crop is also referred to as Ideal Harvesting Time (IHT).

Criteria for harvesting crops

Various crops are harvested at different times or stages. Harvesting at the proper time or stage is crucial to prevent crop loss, possibly due to various reasons, such as too early or late harvesting. These losses may be caused by pests, shattering and other sorts of deterioration. The harvesting of a given crop is determined by various criteria, including:

- (a) Quantity and quality of economic part: The economic part can be the leaf, grain, root, stem, or other parts, depending on the crop type. The crop will be harvested when the desired product is of maximum quality and quantity.
- (b) Intended use of the product: Based on utilisation, the economic part of the product may be the same, but it may be desirable to harvest in different conditions, for example, in a fresh or dry state. Depending on the intended use, crops such as maize, common bean, pigeon pea, and cowpea may be harvested fresh or dry. Thus, the economic purpose of growing the crop determines when is the ideal harvesting time.
- (c) Post-harvesting storage: Most harvested produce, especially legumes and cereal grains, requires drying before storage. Otherwise, if harvested at improper moisture content, the produce may deteriorate in storage facilities.

Methods of harvesting crops

Depending on the crop type, associated cost, and labour availability, several traditional and modern methods can be employed to harvest a produce.

- (a) **Hand harvesting:** Hand harvesting uses simple tools and can have some relative advantages over using machines for example it can be used for different crops with varying maturity times. People can accurately determine product quality, allowing accurate selection of mature produce. This is particularly important for crops with a wide range of maturity and must be harvested several times during the season. However, hand harvesting is labour-intensive and time-consuming.
- (b) **Mechanical harvesting:** This is the modern method of harvesting, for example, using machines like combine harvesters. The advantages of this method are that harvesting is done fast and reduces management problems associated with manual harvesting. However, mechanical harvesting requires that the crop must be grown in a manner that makes mechanical harvesting possible.

Activity 1.10

1. Visit a school farm or any nearby crop field/garden of your choice, then inquire from the farmer on:
 - (a) Maturity indicators of the given crop;
 - (b) Methods, tools and implements used in harvesting the crop; and
 - (c) Challenges associated with harvesting the crop.
2. Through a study visit to any nearby demonstration farm or video clip/ internet search study on:
 - (a) How harvesting practices in large-scale crop production are carried out; and
 - (b) Harvesting implements and technologies used.
3. Use various learning resources such as the library, nearby agricultural extension workers, the internet, and video clips to learn about the different types of digital and computer-based technologies (CBT) or applications used in harvesting.
4. Summarise your observation in your portfolio.

Post-harvesting management

Immediately after the crop's harvestable part is separated from its parent plant or the whole plant is removed from its growing point, it begins to deteriorate. Post-harvesting practises are the activities done to the crop produce at different stages after harvesting. They include those treatments carried out immediately following harvesting (post-harvesting handling) and those that may be done some days or months later. These practices vary depending on various factors, including the nature or type of produce, intended final quality, and the state and time at which the product will be used, stored, or sold. Generally, good post-harvesting handling is critical to reduce post-harvesting losses of fresh produce, maintain quality, preserve nutrient content, and earn higher prices in the market.

Purposes of post-harvesting practices

Generally, post-harvesting handling, among others, aims at:

- (a) Preserving crop products to reduce the rate of deterioration.
- (b) Enable easier handling of the crop produce by reducing their bulkiness.
- (c) Converting crop produce into suitable forms for consumers' use.
- (d) Adding value to the produce.

Depending on these factors, post-harvesting practices may include cleaning, sorting/grading, cooling, processing, packaging, storage and marketing, as discussed herein.

- (i) **Pre-cooling:** Field heat for some crops, especially vegetables and fruits, is usually high and undesirable at harvesting. It should be removed as quickly as possible before any post-harvesting handling activity. Pre-cooling minimises the effect of microbial activity, metabolic activity, respiration rate, and ethylene production. It also reduces the ripening rate, water loss, and decay, thereby preserving quality and extending the shelf life of harvested produce.
- (ii) **Cleaning or disinfection:** Farmers and produce handlers must follow proper cleaning and disinfection practices to prevent post-harvest diseases and reduce the risk of food-borne illnesses being transmitted to consumers.
- (iii) **Sorting and grading:** Sorting involves the removal of rotten, damaged, or diseased fruits and vegetables from the healthy and clean ones. On the other hand, grading is the process of categorising crop produce based on parameters of desirable quality such as colour, size, stage of maturity, or degree of ripening.
- (iv) **Packaging:** This involves packing a crop produce to protect it from mechanical injuries, tampering, and contamination from physical, chemical, and biological sources. Packaging is also essential for efficiently handling the produce into sizeable portions. However, using unsuitable packaging materials can cause product damage, resulting in losses.
- (v) **Storage:** The essence of storage is to extend a product's shelf life and provide continuity of the product supply in the market throughout the year.
Note: The storage conditions, such as temperature and humidity, must be maintained.
- (vi) **Transporting:** Involves the movement of the produce/product from the field to the storage structure or market. Most fresh produce, such as fruits and vegetables, may require specialised refrigerated transportation facilities to avoid deterioration.
- (vii) **Processing:** Processing the food product may involve changing its form into a form that will increase its shelf life. Depending on the nature and form of the food products, processing practices may include drying, canning and freezing. For example, grapes can be processed into wine, cassava into chips, flour and starch and wheat into breads, muffins, pasta, biscuits, cakes, cookies, and pastries.

- (viii) Marketing: Food products can be marketed at farm gate prices on the field, in a local market within the production area, or at distant markets. Marketing of the produce should be considered during planning, even before starting to produce the crop. To maximise profit, it is advisable to market the produce at its highest demand.

Activity 1.11

1. Visit some nearby crop fields and gardens, food markets, and food processing industries. Then, study various post-harvesting handling practices done to the following categories of crops:
 - (a) Roots and tuber crops;
 - (b) Fruits and vegetable crops; and
 - (c) Cereal and pulse crops.
2. Participate in carrying out some of these practices.
3. Select one crop produce and design its suitable packaging.
4. Summarise your observation in your portfolio and share it in class.

Farm record keeping

Farm record keeping is a crucial component of farm management. It involves putting down facts, data or information about the day-to-day activities of a farm. It can also be defined as systematically documenting various farm activities over time. The successful management of a farm requires an accurate, up-to-date record. A complete farm record must include all daily activities and transactions. The farm records guide the farmer in making marketing, pricing and development policies based on costs and returns obtained from farm records.

Activity 1.12

Visit a school farm or some nearby commercial crop farm and perform the following task:

- (a) Observe types of documents used in farm record-keeping
- (b) Inquire on the types of information found on the documents and the importance of each; then
- (c) Prepare a report and present it in class.

Exercise 1.4

1. What criteria are used for sorting and grading of crop produce?
2. How can sorting and grading affect the crop's market value?
3. Citing one crop produce of your choice, suggest and design suitable packaging materials.
4. Propose key information to be recorded at each stage of producing a crop of your choice.
5. Why is record keeping considered an important aspect of farm management?



In your portfolio, summarise what you have learnt from this chapter.

Chapter Two

Banana production

Introduction

In this chapter, you will learn some basic information about banana and its economic importance in daily life. Furthermore, you will learn principles and practices underlying banana production to maximise productivity. The competencies developed will enable you to find beneficial opportunities in the value chain of banana production and apply principles and practices in producing bananas, from planning to post-harvest handling, processing, storage, marketing as well as crop record keeping.



Think

“The giant herb with a false stem that produces fruits which can be eaten raw or cooked.”

Banana and its importance

The banana plant is a large perennial herb with leaf sheaths that form a trunk-like pseudostems. Its scientific name is *Musa* spp. found in the family Musaceae. It is grown for its fruit, which can be eaten either as a ripe dessert or cooked. The banana fruit is mainly rich in carbohydrates, fibres, potassium, vitamins A, B6, and C. Additionally, bananas can be processed into flour, canned slices, jam, jelly, and alcohol. Their widespread popularity makes them an important cash crop. After harvesting the banana bunches, its foliage and pseudostems can be used as animal feed. Bananas are grown on small-scale and large-scale commercial farms (Figure 2.1). The banana industry provides employment for thousands of people worldwide. In Tanzania, the main local banana cultivars are the East African

Highland banana, which includes Matooke, Uganda, Mchare, Kisukari, Mzuzu, Kimalindi, Ndizi ng'ombe, and Bokoboko. On the other hand, improved varieties include TARIBAN 4, TARIBAN 3, TARIBAN 2, and TARIBAN 1.



Figure 2.1 (a): Banana field



Figure 2.1 (b): A bunch of banana fruits

Activity 2.1

Through library and internet searches, perform the following tasks:

1. Identify the major banana-producing areas of Tanzania.
2. Find information about major banana varieties grown in Tanzania and group them according to their respective uses.
3. Record your observations, including lessons learned, in your portfolio.

Principles and practices of banana production

Effective field management of banana crops is crucial for maintaining plant health and ensuring a productive and sustainable banana plantation. The banana field requires careful attention to site selection, good land preparation, selection and preparation of banana planting material, planting at recommended spacing, water, nutrients, pest management, pruning, propping, bagging, and de-suckering.

Choosing a site for banana production

Banana crops grow in deep, well-drained, loamy soils rich in organic matter. It requires an annual rainfall ranging from 1000 - 2000 mm. It can also grow well in altitudes 0 - 2000 m above sea level. The optimal growth temperature and soil

pH ranges for banana production are 20 - 30 °C and 6.0- 7.5, respectively. Banana plants are susceptible to strong winds, which can tear their leaves and cause them to fall over. Therefore, it's essential to protect them by setting up windbreaks. Apart from that, the area should be accessible for easy transportation of banana produce.

Activity 2.2

Through the library, internet searches or inquiries from agricultural experts perform the following tasks:

- (a) Identify the main varieties of bananas grown in your community;
- (b) Find the information on the agro-ecological conditions of your geographical location;
- (c) Comment on the suitability of your agro-ecological conditions for the production of banana; then,
- (d) Summarise your observations and lesson learned in your portfolio.

Land preparation

Before planting, clear bushes and weeds, then deeply cultivate the soil by ploughing and harrowing. The recommended tillage depth is between 30 to 60 cm. After land preparation, prepare planting holes. A 60 cm x 60 cm x 60 cm planting hole is recommended, but in dry and semi-arid areas, use 90 cm x 90 cm x 90 cm holes. Measure and mark the locations for planting holes according to the following spacing: short varieties: 3 m x 3 m (1112 plants/ha), medium varieties: 3 m x 4 m (833 plants/ha), tall varieties: 4 m x 4 m (625 plants/ha). Dig holes by separating topsoil (25 cm) from subsoil. Mix topsoil with manure or compost (20 kg per hole) and 200 g of Triple Super Phosphate (TSP). Add this mixture to the hole first, followed by sub-soil to improve soil fertility and structure. Allow it to settle for at least two weeks before planting/transplanting.

Planting /transplanting

Planting/transplanting a banana plant involves selecting healthy and vigorous planting material before planting or transplanting it into the field.

Selection/preparation of a planting material

Banana is a vegetatively propagated crop; therefore, the planting materials

are tissue culture seedlings and suckers. Tissue culture seedlings (plantlets) are propagated in controlled laboratory environments before being moved to nurseries for hardening-off before sale (refer to Figure 2.2 (b)). The advantages of using tissue culture seedlings include eliminating disease risks (as they are free from diseases), producing a uniform banana crop, and being a fast method of massive planting material multiplication and higher yields. They can be sourced from registered nurseries by the government to sell tissue culture seedlings, such as the Tanzania Agricultural Research Institute (TARI), Sokoine University of Agriculture (SUA), and other reputable agricultural research and training institutes. Most smallholder farmers commonly use sword suckers. They are narrow-leaved shoots about 1 meter tall and have an average of 15 cm diameter (Figure 2.2 (a)) in the lower part. They are sourced from existing banana fields and must be treated with hot water to eliminate nematodes and banana weevils. This process involves the following steps:

- (i) Trimming all the roots;
- (ii) Cutting off 1 cm of tissue around the corm with a sharp knife until clean white tissue is obtained;
- (iii) Prepare hot water; and
- (iv) Dip the trimmed corm in the hot water bath at 50 - 55 °C for 20 minutes. Under a field situation where there is no thermometer, the suckers can be dipped in boiling water for 30 seconds.



Figure 2.2 (a): Sword suckers

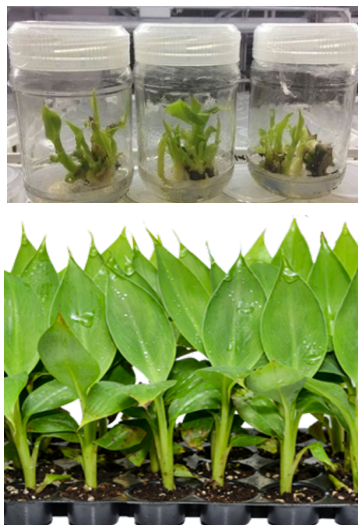


Figure 2.2 (b): Tissue culture banana seedlings

Source: https://agritech.tnau.ac.in/bio-tech/biotech_tc_jainirrigation.html

Planting/transplanting

The sucker is placed in the planting hole, and its corm is covered first with the topsoil and then the subsoil. For tissue culture planting material, transplanting should be performed when tissue culture seedlings are approximately 30 cm tall and have developed at least five healthy leaves. For proper anchorage, each sucker or seedling should be planted 30 cm deep in the centre of a planting hole. Remove the polythene bag carefully (for tissue culture seedlings). Cover the roots with soil and ensure the plant is upright and firm. After transplanting banana seedlings, apply a thick layer of mulch around each plant to conserve soil moisture. In rain-fed conditions, planting should be done at the start of the rainy season. However, if irrigation is available, planting can be done year-round.

Activity 2.3

1. Visit your school farm and nearby banana fields and perform the following tasks:
 - (a) Select a suitable area in the school farm and prepare a hole for planting banana suckers according to the recommendations;
 - (b) Identify different types of banana suckers;
 - (c) Select a suitable banana sucker that can be used as a planting material;
 - (d) Prepare the holes as per the recommended spacing; then,
 - (e) Plant the selected suckers and manage them.
2. In your portfolio, design a form specifying the types of records/information you will need to keep in the production of banana from planning to marketing of the crop produce/ products.

Exercise 2.1

1. Explain why farmers are advised to use clean and certified banana planting materials.
2. Why should the polythene bags be removed from tissue culture banana seedlings during transplanting?
3. Explain how you can reduce moisture stress in transplanted banana seedlings.
4. Not all banana suckers are suitable for planting. Discuss.

Soil water management

Banana plants require a minimum of 1,000 mm of rainfall per year. Therefore, at all times of plant growth, soil water management should aim at maintaining soil water content at an optimal state for banana plant growth and productivity. If the annual rainfall is less than 1,000 mm, the provision of water becomes necessary, especially during the flowering stage. Similarly, excess soil water is not good for the plant and hence has to be drained off. In drier seasons, supplemental water is even crucial. This can be done through various means of irrigation including flooding and drip irrigation (Figure 2.3). Cultural practices such as the use of mulches around banana plants, and applying organic matter minimises the needs for irrigation.



Figure 2.3: Drip irrigation system in a banana field

Activity 2.4

1. Adopt a suitable soil water management system and apply it to the planted banana suckers (refer to Activity 2.3);
2. Explain the advantages and limitations of the system; then,
3. Make a summary.

Exercise 2.3

1. What criteria would you use to recommend on a suitable irrigation system in the banana field?
2. How do mulches and organic matters facilitate water management in the banana field? Give three points.
3. Explain how can you manage water logging in the banana field?
4. With reasons, suggest a suitable water management method of your choice for the banana field.

Nutrient management

Bananas require specific nutrients for optimal growth, fruit development, and high yields. The essential nutrients for banana plants include nitrogen (N), phosphorus (P), potassium (K), secondary nutrients like magnesium (Mg), calcium (Ca) and sulphur (S), as well as micronutrients such as zinc (Zn), iron (Fe), manganese (Mn), copper (Cu), boron (B), and molybdenum (Mo).

Banana plants require a high concentration of macronutrients (nitrogen (N), phosphorus (P), and potassium (K)) due to their significant nutrient needs. Fertilisers with higher NPK ratios (e.g., 20-20-20) are preferred for bananas, though a balanced NPK ratio like 8-10-10 or 15-15-15 is recommended. Alternatively, CAN fertilisers can be used.

Fertilisers should be applied twice a year during the rainy season. Apply it per plant, placing the top-dressing fertiliser in a band 60 cm away from the plant. Alternatively, use an NPK fertiliser rich in potassium at 250 g per plant. Additionally, apply 2 to 4 times annually of 20 – 40 kg of well-decomposed farmyard manure (FYM) per stem before the onset of rains. This manure should be spread around the outer diameter of the canopy and incorporated into the soil carefully to avoid damaging the roots (Figure 2.4). Generally, the amount of fertiliser applied depends on factors such as environment, variety, age, and size of banana crop, soil nutrient status and the amount of plant nutrients depleted per year.

Procedure for fertiliser application in banana

Before applying fertilisers to any crop, including bananas, it is important to consult agricultural experts to conduct soil tests to determine the status of nutrients and pH level for the given crop. This helps understand what nutrients are lacking and the type of fertiliser required. The following procedure is recommended when applying fertiliser to banana plants.

- (i) Weed the field;
- (ii) Choose the right fertiliser;
- (iii) Water the plant a day before applying fertiliser to help with absorption;
- (iv) Spread the correct quantity of fertiliser evenly around the base, but ensure it does not touch the stem;
- (v) Water the plants thoroughly after applying the fertiliser to help dissolve the nutrients and carry them to the roots;
- (vi) Apply mulch around the base of the plant to retain moisture and suppress weeds. Use organic materials like straw, grass clippings, or leaves; and
- (vii) Keep records of fertilisation schedules, types, and amounts used for future reference and to track plant performance.



Figure 2.4: Fertiliser application in banana

Pest management

Banana crop is affected by several pests. The major pests affecting the crop include weeds, nematodes and diseases. These should be dealt with to maximise crop productivity. Management practises for each are explained herein.

Weed management

Banana fields should be maintained free of weeds by hoeing, mulching or using herbicides. Bananas have shallow roots, so caution is necessary during weeding to prevent root damage. A thick layer of mulch (10-15 cm) can effectively suppress weeds, retain moisture and contribute humus to enhance soil structure. Additionally, inter-cropping can help to keep the field free of weeds. Consult an agricultural extension worker for the appropriate herbicides, correct dose and time and how to apply them to crop plants.

Activity 2.5

Visit some nearby agrovet shops or agricultural extension workers and do the following:

- (a) Inquire about the types of fertilisers recommended for banana plants in a field;
- (b) Identify the types of weeds that affect banana production;
- (c) Identify the available herbicides for controlling weeds in banana fields;
- (d) For each type of herbicide, record the recommended dose; then,
- (e) Summarise your observations in your portfolio.

Exercise 2.4

1. Describe different techniques that help to manage soil moisture in banana fields.
2. What are the primary nutrients required by banana plants? How can they be supplied efficiently?
3. Outline the major pests affecting a banana and suggest ways to manage them.
4. Explain what precautions are to be considered when applying pesticides on the banana field.

Diseases management

Bananas are susceptible to various diseases. Some of the major banana diseases include:

- (i) **Banana Bacterial Wilt:** The disease is caused by the pathogen *Ralstonia solanacearum*. It spreads easily through infected planting materials, farm tools, and insects, including bees, through the male bud. Symptoms of the disease include wilting of leaves, yellowing, and bacterial ooze from cut stems. It is controlled by Quarantine measures, destruction of infected plants, disbudding of male flower buds after fruiting, and soil treatment.
- (ii) **Bunchy Top Disease:** This disease is caused by *Banana bunchy top virus* (BBTV) and transmitted by banana aphids at any growth stage. It can also be spread through infected planting material such as suckers and tissue culture plants.

Symptoms: The disease starts as dark green spots or streaks on minor veins or the midrib. Later leaves become smaller, chlorotic with upside-down edges. They become dry, brittle and erect, creating a “bunchy top” look in banana plants (Figure 2.5). Flowering is limited, and only very small bunches may be produced from infected plants.

Management: The disease can be managed by controlling banana aphids, removing and destroying infected plants, and using virus-free banana planting materials.



Figure 2.5: Symptoms of Bunchy top disease in banana plants

(iii) Cigar-end Rot: This disease is caused by the fungus *Verticillium theobromae*.

Symptoms: The disease starts at the tip of the banana, showing up as small, dark brown to black spots. These spots enlarge and expand towards the base, causing the tip to shrivel and darken like a burnt cigar end, hence the name “cigar end rot” (Figure 2.6). In severe cases, the rot can penetrate deeper into the fruit, affecting the flesh and making the banana fruits unsuitable for consumption. It is favoured by high humidity caused by overcrowded fields or stools and abundant leaf trash.

Management: The disease can be managed by ensuring proper field hygiene and eliminating excess suckers. It is also recommended to remove male flower buds that are located 15 cm below the last hand once bunch formation is complete. You can consult agricultural experts for advice on the appropriate fungicides to use.



Figure 2.6: Symptoms of Cigar-end Rot in banana fruits

(iv) Black Sigatoka: The disease is caused by the fungus *Mycosphaerella fijiensis*.

Symptoms: The disease begins as small, dark streaks on the undersides of the leaves. These streaks expand and merge into large black or dark brown patches, sometimes with a yellow halo. Infected leaves turn yellow, wither, and die early, reducing photosynthesis and stunting plant growth (Figure 2.7).

Management: To manage the disease, implement cultural practices like removing infected leaves, ensuring proper plant spacing, and maintaining efficient drainage in fields. Resistant cultivars such as the TARIBAN series, FHIA 17, 18, 25, Williams, and Grand Nain are recommended. Overhead irrigation is not recommended because it favours the disease. You can consult agricultural experts for advice on the appropriate fungicides to use.



Figure 2.7: Symptoms of Black Sigatoka in banana leaf

- (v) **Panama disease:** Panama disease, also known as Fusarium wilt is caused by the soil-borne fungus *Fusarium.spp*.

Management: The disease can be managed using resistant varieties, such as Giant Cavendish or FHIA hybrids. It is recommended to use disease-free material (Tissue Culture and clean suckers). Other management measures include restricting movement of plant material from infected areas to non-infected areas (quarantine).



Figure 2.8: Symptoms of Panama disease in banana plants

(vi) **Anthraco***nose* This is a fungal disease that affects banana. It is caused by *Colletotrichum musae*.

Symptoms: Symptoms involve dark, sunken spots on the fruit that may develop into a soft rot over time. In more severe cases, the disease can also affect the leaves and pseudostems. The fungus produces pinkish (Figure 2.9). The fruit's flesh remains unaffected unless it becomes overripe.

Management: The disease can be managed by removing and destroying infected plant material, ensuring proper spacing between plants for good air circulation and practicing crop rotation. Minimising bruising in post-harvest handling operations and hot water treatment (50°C for 5 minutes) on fruits are also recommended.



Figure 2.9: Symptoms of Anthracnose disease in banana fruit and leaf

Vermin management

In some areas especially those close to the forestry large animals (vermin) such as monkeys and wild pigs can cause significance damage to the banana crop.

The following are some practices that can be employed to manage them:

- (i) Ensuring that banana fields are established away from forested areas;
- (ii) Implementing controlled hunting;
- (iii) Aligning the crop plant in blocks;
- (iv) Using a technique of capturing one monkey, and painting with a striking colour or hanging it with a bell around its neck and release it to threaten others.

Activity 2.6

1. Pay a visit to your school farm or nearby banana fields observe and perform the following tasks:
 - (a) Identify some common pests and diseases affecting banana crop;
 - (b) Study methods that are used to manage pests and diseases in the banana fields; then,
 - (c) Adopt management measures that are applicable in your school farm.
2. Based on your experience in your locality, suggest other techniques that are used in managing vermin in banana fields.

Exercise 2.5

1. Define the following terms as used in banana:
 - (a) Bunch;
 - (b) Hand;
 - (c) Finger;
 - (d) mat; and
 - (e) male bud.
2. Outline equipment and materials that are required for harvesting bananas.
3. How can banana bunches be protected from insect damage in the field?
4. Briefly prepare a summary on Black sigatoka disease in banana to be presented in one of a group of banana producers in your area.
5. Why is it important to harvest bananas at the right stage of maturity?

Other agronomic practices in banana production

There are other management practices which are specific to a banana crop. These include de-suckering, propping, de-leafing, de-budding and bagging.

De-suckering

De-suckering in banana implies removing unwanted suckers from stools (Figure 2.10). The importance of which is to limit nutrient competition amongst banana suckers and to improve air circulation within the canopy. To manage unwanted suckers, ensure each stool has only three suckers: a mother plant (bearing sucker) and a daughter plant (half-grown sucker). and a granddaughter plant (sprouting sucker). Remove excess suckers with their corms, and you can use them as

planting materials. To prevent sprouting, insert a peg into the growing point. Start this process two months after planting and repeat every 45 days until the plant flowers.



Figure 2.10: De-suckering in a banana field

Propping

Propping supports banana plants with fruit bunches to prevent them from falling. A pole is placed under the bunch, taking care not to damage the fruit (Figure 2.11). Tall banana varieties, such as Grand Nain, Williams, Valery, Giant Cavendish, and the FHIA series, require propping.



Figure 2.11: Propping in banana

De-leafing

De-leafing in banana plants involves removing old, damaged, or diseased leaves to improve plant growth and health. This process improves photosynthesis by allowing better sunlight penetration to healthy leaves and fruit bunches. It also prevents the spread of fungal and bacterial infections.

De-budding/de-navelling

This refers to removing the male bud from a banana plant after the bunch formation is complete. Tools used during this process should be disinfected. The benefits of removing the male bud include increased yield, uniform hands, and reduced thrips attack.

Bagging

Covering banana bunches with special plastic bags protects them from thrips and other insects and speeds up the maturation process.



Figure 2.12: Bagging in banana

Harvesting

Bananas should be harvested at the right stage of maturity in order to get good-quality fruits. It is recommended to follow the maturity indices such as fruit size, shape, peel and flesh colour to achieve the purpose. The maturity indices are as elaborated hereafter.

- (a) Fruit size: As fruit matures, both its length and volume increase.
- (b) Fruit shape: At the initial stage of development, individual fruits (fingers) have an angular cross-section. As they mature, the fruits become more rounded.
- (c) Peel and pulp colour: As the fruit matures, the peel colour changes from deep green to light green or yellow, while the pulp colour changes from cream to orange-yellow.

Harvesting involves cutting the bunch from the pseudostem. For taller varieties, it may be necessary to cut the pseudostem halfway to access the bunch and prevent it from falling. It begins 9 to 18 months after planting. Bananas reach full production in 2 to 3 years. On average, banana yields in Tanzania range from 10 to 15 tons per hectare per year. However, with improved agricultural techniques and better varieties, higher banana yields can be attained.

Activity 2.7

Pay a visit to your school farm or family banana field, then, under permission of your teacher/parents or guardian, perform the following tasks:

1. Observe and identify the maturity indices of the banana bunches;
2. Participate in selecting and harvesting some matured banana bunches for home/school use; then,
3. Summarise your observations in your portfolio.

Exercise 2.6

1. Explain what does harvesting implies in banana.
2. Why it is important to harvest banana at the right stage of maturity?
3. Outline common equipment and materials that are required for harvesting bananas.
4. Describe the proper way of harvesting both the short and long banana varieties in the field.

Post-harvest management of banana

Post-harvest handling of bananas focuses on ensuring quality and reducing losses. Therefore, harvested bunches should be handled gently to avoid bruising. Farmers normally transport and market their bananas in bunches, which can expose the fruits to mechanical damage and reduce their quality. To minimise this damage, it is recommended to remove the individual hands from the bunches and pack them in reusable plastic containers.

The following post-harvest handling practices are essential to ensure high-quality bananas.

- (a) **Cleaning:** Banana should be cleaned to remove dirt and debris, but avoid using excessive water which can lead to spoilage.
- (b) **Sorting and grading:** It is advisable to sort bananas based on criteria such as free from mechanical damage, foreign matter, decay, freshness, maturity and latex staining. Grade them by size and quality. Remove any damaged or overripe bananas to prevent them from affecting the rest.
- (c) **Packaging:** It is recommended to pack bananas in well-ventilated containers to prevent bruising and ensure proper air circulation. Use materials that provide cushioning.
- (d) **Ripening:** If bananas need to be ripened, follow the following steps:
 - (i) Choose mature, green bananas. Ensure they are free from bruises and defects;
 - (ii) Clean the bananas to remove dirt and debris. Sort the bananas by size and quality to ensure uniform ripening;
 - (iii) Store the bananas in a ripening room (about 17 °C-20 °C);
 - (iv) Cover them with the materials that allow air to go through;
 - (v) Monitor the ripening process regularly;
 - (vi) Pack the bananas carefully to avoid bruising during transportation and handling; then,
 - (vii) Distribute the ripe bananas to retailers or sell them directly to consumers.
- (e) **Processing:** Bananas can be used to make many products. These include wine, flour, juices, chips/crisps, jams/jelly, and sweets. Un processed banana can be stored at cool temperature (around 13 °C-14 °C to slow ripening, prevent spoilage and extend their shelf life.

(f) Marketing: Understanding market aspects can help banana stakeholders benefit from opportunities in the banana production industry. Some of these marketing aspects are as follows:

- (i) Branding and packaging: Develop a strong brand and use durable packaging that attract buyers.
- (ii) Market segmentation: Identify and target different market segments, that is local market such as rural/urban markets, restaurants, and supermarkets; and export markets. It is important to plan and design an approach to meet the specific needs of each segment. Moreover, sanitary and phytosanitary standards and measures for the crop produce must be taken into consideration especially when targeting export markets.
- (iii) Quality assurance: Because different markets have different quality standards, it is advisable to implement proper post-harvest handling practices, such as cleaning, sorting, and packaging.
- (iv) Partnerships: Partner with local retailers, supermarkets, and restaurants to supply them with bananas. Establishing long-term contracts can ensure a steady demand.
- (v) Promotions and discounts: To attract more customers, run promotions, offer discounts, or create bundles. For example, offer a discount on bulk purchases or during special events.
- (vi) Value-added products: Diversify your product range by offering value-added banana products such as banana chips, dried bananas, or banana-based snacks and desserts to capture more customers.
- (vii) Customer engagement: Engage with customers through social media. Post banana images on your page, including details such as location and selling price, newsletters, and in-store events. Share recipes, health benefits of bananas, and updates about your farm or business.
- (viii) Seasonal strategies: Capitalise on the peak banana seasons with targeted marketing campaigns and ensure consistent supply throughout the year to maintain customer trust.
- (ix) Export opportunities: It is recommended that export markets be explored and that export regulations be complied with, to reach a large.

Activity 2.8

1. Visit a banana field at your school or neighbouring farms and perform the following tasks:
 - (a) Identify the banana variety suitable for ripening;
 - (b) Identify its proper maturity stage according to the maturity indices;
 - (c) Prepare the place for ripening and follow the ripening procedure; then,
 - (d) Design the marketing strategy for the ripe banana.
2. Record your observations and experience during the activity, including the steps you took, the challenges you faced, and the lessons you learned.
3. Use the collected records/information you have been recording in all stages of banana production to:
 - (a) Evaluate the profitability of banana production enterprise; and
 - (b) Suggest measures to improve the enterprise and share in class.



In your portfolio, write a summary of what you have learnt from this chapter.

Chapter Three

Cassava production

Introduction

In this chapter, you will learn the principles and practices of cassava production. The competencies developed will enable you to produce, process, store, market and keep records of cassava production. Also, you will explore various opportunities available in the value chain of cassava production.



Think

“The hunger fighter with underground hidden edible treasure and edible leaves.”

Cassava and its importance

Cassava (*Manihot esculenta*) is an essential annual crop native to South America. It is propagated from stem cuttings since the tubers do not produce buds. Its starchy, thickened tuberous roots provide an affordable source of calories, especially in areas where calorie deficiency and malnutrition are common. Other nutrients found in cassava are fibre, vitamin C, thiamine, folate, vitamin B6, potassium, magnesium, and niacin. The major production regions in Tanzania include Mwanza, Mtwara, Lindi, Shinyanga, Tanga, Ruvuma, Mara, Kigoma, Morogoro, Pwani, and most areas in Zanzibar.

Moreover, cassava can be processed into flour, chips, canned goods, and alcohol. Its widespread popularity makes it a vital subsistence food crop and a cash crop for smallholder farmers. The plant's foliage is also used as a vegetable. In Tanzania, both local and improved varieties are cultivated. Improved varieties include Mkombozi, Kiroba, Kizimbani, Mkumba, Kipusa, Mkuranga 1, and the TARICASS series.



Figure 3.1 (a): a Field of cassava

Source:<https://www.linkedin.com/pulse/unlocking-potential-cassava-smart-crop-kenyas-asal-regions-khasoa-l9vwf>



Figure 3.1 (b) Cassava root tubers

Source:https://www.iita.org/research/our-research-themes/improving-crops/1024_cassava-tubers_opt/

Exercise 3.1

1. Describe the main uses of cassava crop.
2. Describe how cassava contributes to food security in Tanzania.
3. Why is cassava considered an essential staple food in many developing countries?
4. Mr. Mpenda Kazi's cattle ate cassava from Mrs. Kilimo Kwanza's farm. Shortly after, the cattle exhibited the following symptoms: rapid breathing and excessive salivation. Mr Mpenda Kazi accused Mrs Kilimo Kwanza of bewitching his cattle. However, the examination report from a livestock officer revealed that there were a factor relating to cassava behind this scenario. Use an internet or library search to find out the possible main cause from cassava for this situation to resolve the dispute.

Principles and practices of cassava production

Effective cassava production ensures high yields by correctly choosing the production site, selecting cassava planting materials, planting, water management, nutrient management, pest management, harvesting, and post-harvest management.

Choosing a site for cassava production

Adhere to a crop's ecological requirements for the successful production of cassava. Ecological factors include soil factors (soil pH, soil type, and fertility) and climatic and physical factors such as rainfall, temperature, altitude, and topography.

It grows best in tropical and subtropical climates with temperatures between 25-29°C and altitudes between 0-1,500 m above sea level (a.s.l). Cassava requires an annual rainfall between 1,000 to 1,500 mm. The crop can tolerate lower rainfall but performs best with consistent moisture throughout the growing season. A well-drained sandy loam soil with a pH of 5.5-7.0 is optimum for cassava production. Waterlogged or saline soil is not recommended for cassava production. Mechanical measures like contouring should be implemented on steep slopes to prevent soil erosion.

Land preparation for cassava production

In cassava production, land preparation involves clearing the site, ploughing, and making ridges or mounds for planting. Land preparation can be done using a hand hoe, motorised tiller, or tractor-drawn instruments to plough the soil and form ridges. Ploughing the soil at a 20-30 cm depth is recommended to ensure it is loose enough for the roots to develop properly. The land should be cleared of weeds, shrubs, and old crop residues.

Activity 3.1

Visit the school farm and then perform the following tasks:

- Identify the suitable site for cassava crop establishment;
- Identify equipment that can be used for land preparation;
- Use the identified equipment to prepare the land for cassava planting; and
- Make ridges at the appropriate spacing.

Selection of planting materials

Cassava is vegetatively propagated using stem cuttings. Vegetative multiplication is a simple approach with the advantage of producing vigorous plants. Many improved cassava varieties are suited to different agroecological zones and production purposes. Choosing a marketable variety and disease-resistant variety is recommended for commercial and subsistence cassava production. Consult agricultural extension workers and experienced farmers for available cassava varieties that are resistant to disease in the targeted area. Choose healthy, disease-free cassava plants 8-12 months old with mature stems 2-3 cm in diameter for cutting. After choosing the right cassava plants, remove the top and bottom ends.

Cuttings should be stored in a cool, shaded environment for 2-3 weeks. Cuttings can be prepared using a sharp knife at about 20-30 cm long. (refer to Figure 3.2)



Figure 3.2: Cassava stem cuttings

Source: <https://www.amazon.in/Creative-Farmer-Esculenta-Cuttings-Planting/dp/B09QSKVBJL>

Planting cassava

Cassava planting can be done either manually or using a machine. In Tanzania, the planting process is mainly done manually. Cuttings can be planted on ridges, mounds, or flat ground. To plant cassava, follow the following steps:

- (i) Prepare ridges or mounds or flat land for planting cassava. Hole spacing should be 1 meter between plants and 1 meter between rows (1m x 1m), which gives a population of 10,000 cassava plants (Figure 3.3 (a)). Pick the prepared cuttings of about 20-30 cm long.
- (ii) Insert the cuttings about the centre of a ridge/mound at a depth of 5-10 cm into the soil. Ensure cuttings are positioned at a 45-degree angle with at least two nodes buried in the soil and two above the ground.
- (iii) Water the cuttings immediately after planting if the soil is dry.

The cuttings will take about 1-2 weeks to sprout after planting (refer to Figure 3.3 (b))



Figure 3.3 (a): Planted cassava cuttings



Figure 3.3 (b): A sprouting cassava cutting

Sources: <https://www.shutterstock.com/image-photo/cassava-tapioca-plants-manioc-germinate-branch-1033869577>

Activity 3.2

1. Visit your school farm or nearby cassava fields and perform the following tasks:
 - (a) Select a suitable cassava cutting for planting;
 - (b) Plant the cuttings in ridges according to the recommendations; then,
 - (c) Manage the planted cuttings and keep the record in your portfolio.

Exercise 3.2

1. Explain why harrowing is an essential activity in land preparation.
2. Mention types of herbicides used in land clearing.
3. Why is it necessary to carefully select cassava planting materials?
4. Mr. and Mrs. Mhogo-Mali prepared cassava cutting seeds correctly for planting. They timed the cropping season and planted the cuttings at the correct spacing. The cuttings were expected to sprout after a week since were planted. Unfortunately, when the couple visited the field, they found the cutting did not sprout as expected. Explain the possible reasons why the cuttings could not sprout.

Soil water management

Cassava can tolerate drought. However, it requires sufficient moisture in its early growth stage. Optimal growth occurs with 1,000 mm or more of annual rainfall. Proper soil moisture during establishment boosts yields. Cassava yields can improve significantly when supplement water is supplied to crop plants through various means, including drip irrigation (Figure 3.4). However, excessive irrigation can lead to waterlogging conditions and cause root rot and stunted growth. Therefore, appropriate soil water management practices are necessary to keep soil moisture content at optimal levels.



Figure 3.4: Drip irrigation

Source: <https://elemental.green/drip-irrigation-systems-the-water-saving-solution/>:

Activity 3.3

Visit the established cassava plot in the school field/nearby cassava fields and perform the following tasks:

- (a) Observe water management practices employed in the fields;
- (b) Adopt any suitable management practice and apply it in the cassava plot you have established; then,
- (c) Summarise the lesson learnt in your portfolio.

Nutrient management

Adding nutrients to the soil helps to maintain soil health and enhances depleted or poor soils. Both organic and inorganic fertilisers can be used in cassava production. Organic manures, such as poultry manure, cow dung, and compost, supply essential nutrients and organic carbon. Organic manures improve the soil's physical properties, water retention, and microbial activity, resulting in fertile soil.

Using chemical fertilisers, smallholder farmers can increase their cassava yields from about 10 to 16 tonnes of fresh roots per hectare. In Tanzania, most farmers do not apply fertiliser to their farms. It is recommended to use the following four key principles when using fertiliser in cassava production:

- (a) the correct type of fertiliser;
- (b) right rate of application;
- (c) right timing of application; and
- (d) the correct method of application.

A compound fertiliser with the composition of nitrogen (N), phosphorus (P), and potassium (K), alongside with micronutrients such as boron and zinc, is advised for increasing cassava yields. Other fertilisers such as urea, single superphosphate (SSP) or triple superphosphate (TSP) and potassium chloride (KCl, also known as muriate of potash) can be used. The amount of fertiliser to be applied should be based on soil test results and applied in three phases: One-third of the total quantity should be applied 4-6 weeks after planting. The second application should be done 10–12 weeks after planting, followed by a third application 16–20 weeks after planting. Consult agricultural experts about the type, rate, timing, and method of fertiliser application based on the nutrient status of the field soil.

Activity 3.4

1. Through farm visits, market survey, library, and internet search:
 - (a) Identify different types of organic and inorganic fertilisers that are recommended for cassava production; and
 - (b) Indicate the amount and the right time of application for each.
2. Under the guidance of your teacher, choose the right fertiliser and apply it to the cassava plants you have established.
3. Keep observing and recording the progress of the crop plants in your portfolio.

Pest management

Cassava is affected by various pests that reduce its yield and quality. Significant cassava pests include weeds, viruses, cassava mealybugs, green mites, variegated grasshoppers, and whiteflies. Effective management strategies against these pests can improve productivity and quality.

Weeds management

Cassava is affected by annual and perennial weeds (Figure 3.5 (a)). Weeds can compromise growth and reduce cassava productivity. Although cassava yields vary with cultivars, season of planting, soil type and fertility, with improved varieties and under good management practices, they can reach 20–25 tonnes per hectare. Poor weed management can cause yield loss of 30% to 40%. The frequency of weeding cassava fields depends on the type, amount, and severity of the weed problem. It is recommended that the first weed control be performed 3–4 weeks after planting. It should be repeated at 8 and 12 weeks, and the final weeding should be done between 20 and 24 weeks after planting (Figure 3.5 (b)).



Figure 3.5 (a): Cassava field infested with weeds



Figure 3.5 (b): Clean cassava field

Source: <https://cassavamatters.org/cassava-weeds/>

Insect pest management

Effective insect pest management involves strategies to control pests that can harm crop yield and quality. Fundamental approaches include biological, cultural, mechanical, and chemical methods. Cultural control involves crop rotation and sanitation, while biological control uses natural predators and pathogens. Mechanical methods include traps and barriers, and chemical control involves careful, targeted insecticide use.

(a) Cassava mealybug: These are small, sap-sucking insects that infest cassava plants. They are pink, soft-bodied insects covered with a white, waxy cotton material (Figure 3.6)

Management of cassava mealybug: The insect is managed using planting materials free from cassava mealybug infestation, destroying severely infested plants by burning them, and using insecticidal soap mixed with oil or as per agricultural expert's recommendation.



Figure 3.6: Cassava mealy bug

Source: <https://plantwisepiusknowledgebank.org/doi/10.1079/PWKB.20177801289>

(b) Cassava green mites: They are tiny, greenish-yellow pests that infest cassava plants, feeding on the undersides of leaves (Figure 3.7 (a) & (b)).

Symptoms: Yellowing, mottling, death, and leaf fall.

Management: Plant early at the onset of rain because their effect is severe during the dry season, practising crop rotation. Spray recommended insecticide.



Figure 3.7 (a): Cassava green mite



Figure 3.7 (b): Cassava green mite damage symptoms on cassava leaf

Source: <https://plantwiseplusknowledgebank.org/doi/10.1079/PWKB.20167800978>

(c) Whiteflies: They are small, sap-sucking insects (Figure 3.8).

Symptoms: They suck sap from cassava leaves, leading to yellowing of leaves, wilting, and stunting of the plant. They excrete honeydew, a sugary substance that promotes the growth of sooty mould.

Management: Whiteflies are managed by removing and destroying infested plant debris to prevent the spread of whiteflies. Rotating cassava with non-host crops and practising intercropping with plants that repel whiteflies, such as marigolds, is advisable. Insecticides can be used under the guidance of agricultural experts.



Figure 3.8: Whiteflies on cassava leaf

Source: <https://www.mpg.de/15470351/whiteflies-detoxify-plant-defenses>

- (d) **Variegated grasshopper:** A striking insect known for its vivid colour (Figure 3.9). When threatened by an enemy, it produces an unpleasant smell as a defensive mechanism.

Symptoms: Cassava leaves may have irregular holes or chewed edges due to the grasshoppers' feeding habits.

Management: Variegated grasshoppers can be managed by removing weeds and plant debris around cassava fields to reduce grasshopper breeding sites. Some farmers control pests manually by removing them from cassava plants and killing them. Intercropping or planting crops that are less attractive to them can help to manage the disease. It is recommended to consult agricultural experts for appropriate insecticides against variegated grasshoppers.



Figure 3.9: Variegated grasshopper on cassava leaf

Source: <https://plantwiseplusknowledgebank.org/doi/10.1079/pwkb.20167800867>

Disease management

Cassava is susceptible to various diseases that result in reduced yield and lower quality of its tubers. The most harmful are viral infections, particularly Cassava brown streak disease (CBSD) (Figure 3.10) caused by the *Cassava brown streak virus* (CBSV) and Cassava mosaic disease (CMD) caused by the *Cassava mosaic virus* (CMV) (Figure 3.11).

Management: The disease can be managed by inspecting cassava fields and removing (roguing) infected plants to prevent the virus from spreading to healthy plants. Also, cleaning working tools and equipment to avoid mechanical transmission of the virus. Implementing quarantine measures to prevent the introduction of infected planting material from other regions. Crop rotation and controlling aphids. Planting during periods less favourable for whitefly vectors can help reduce infection rates.



Figure 3.10: Cassava brown streak disease symptoms

Source: <https://www.plantsci.cam.ac.uk/news/developing-predictive-model-emerging-epidemic-cassava-sub-saharan-africa>



Figure 3.11: Cassava Mosaic disease symptoms

Source: <https://plantwiseplusknowledgebank.org/doi/10.1079/PWKB.2014780137>

Activity 3.5

1. Pay a visit to your school farm or nearby cassava, banana, or sweet potato crop fields, observe and perform the following tasks:
 - (a) Identify some common vermin affecting the crops;
 - (b) Identify methods that are employed to manage vermin in the studied crop fields; and
 - (c) Adopt manageable measures for you and apply them in controlling vermin in the school crop fields.
2. Summarise your observations in your portfolio.

Exercise 3.3

1. What factors should be considered when recommending the application of fertilisers in cassava fields?
2. Find out other cassava diseases apart from CMD and CBSD.
3. Can cassava be intercropped with other crops? Explain.

Harvesting in cassava

Harvesting cassava can be done mechanically using cassava harvesting machines or manually using hand tools like a hoe, spade, and folk. In Tanzania, cassava roots are usually harvested by hand. This process is easier when the soil is moist. The following are steps to be followed when harvesting cassava manually;

- (i) Cut the upper parts of the stems leaving about 30-50 cm height;
- (ii) Use a hoe, spade, or fork to loosen the soil;
- (iii) Pull out the roots by hand (Figure 3.12 (a));
- (iv) Remove soil and any remaining dirt from the roots. Avoid washing them, as moisture can cause spoilage;
- (v) Check the base of the cassava for any broken-off roots and dig for them out of the soil; and,
- (vi) Use a clean sharp knife to trim the root tubers from the stems.

Cassava harvesting can be semi-mechanised or fully mechanised and involves several steps, including loosening the roots, pulling up the crops, removing the soil, and separating the roots (Figure 3.12 (b)). The roots are then collected and loaded for transportation.



(a)



(b)

Figure 3.12: Harvesting in cassava; (a) manually pulled-off the ground (b) mechanised harvesting

Post-harvesting management

Cassava tubers should be processed or consumed relatively quickly after harvesting, as they can spoil rapidly. If not used immediately, store them in a cool, dry, well-ventilated place. For extended storage, cassava can be processed into flour, chips, or other products. Tubers can also be peeled, sliced, and dried in the sun or using a mechanical dryer. Alternatively, cassava can be fermented to produce more extended shelf-life products.

Marketing cassava

Marketing cassava can involve different strategies to ensure it effectively reaches the target market. Some of these strategies include:

- (i) Identifying the target market: Local consumers or companies that use cassava for products like starch, flour, and animal feed.
- (ii) Diversifying product: Process cassava into products such as cassava flour and cassava chips.
- (iii) Branding and packaging: Develop a strong, recognisable brand. Invest in attractive, informative, and sustainable packaging that appeals to consumers.
- (iv) Identifying marketing channels: Use social media and a dedicated website to reach a wider audience. Place your products in supermarkets and speciality food shops, and partner with wholesalers to distribute your products to various retail outlets.
- (v) Promotion strategies: Use traditional media (TV, radio) and digital marketing (social media). It is advisable to offer sales discounts and special offers to attract and retain customers.

Activity 3.6

1. Visit the nearby market to assess the following on cassava and cassava products.
 - (a) Where does the cassava produce come from?
 - (b) Identify the cassava products found in the market.
 - (c) Identify the price of cassava produce and its products.
 - (d) Ask how cassava is stored to increase its shelf life and how long it takes to deteriorate.
 - (e) Summarise your observations in your portfolio.

2. Based on the information you have been recording at various stages of cassava production, assess success, challenges and the profitability of cassava production.

Exercise 3.4

1. How can you know the right time to harvest cassava?
2. Suggest the promotion strategies that can be used to market cassava crops.
3. What type of transport is used to move cassava from the field to the market?
4. How can you increase shelf-life and maintain the quality of harvested cassava roots?



In your portfolio, summarise what you have learnt from this chapter.

Chapter Four

Sweet potato production

Introduction

In this chapter, you will learn the principles and practices of sweet potato production. The competencies developed will enable you to produce, process, store, market and keep records of the production of the crop. You will also explore available opportunities in the value chain of sweet potato production.



Think

A humble crop, vital for sustenance and resilience.

Sweet potato and its importance

Sweet potato (*Ipomoea batatas*) is a root crop primarily cultivated for its starchy tubers, though its leaves are also edible. It is rich in vitamins, minerals, and beta-carotene. It can be orange, white, or purple (Figure 4.1(b)). It is produced throughout Tanzania for market in both rural and urban areas. The major producing regions in Tanzania include Shinyanga, Mara, Mwanza, Kagera, Kigoma, Tabora, Morogoro, and Mbeya. Both local and improved varieties are grown in Tanzania. Improved varieties include Kabode, Kakamega, Mataya, Kiegea, Ejumla/Mlezi, Polista, and Ukerewe.



Figure 4.1(a): Sweet potato plants

Source: <https://gardeningsg.nparks.gov.sg/page-index/edible-plants/sweet-potato/>



Figure 4.1(b): Potato roots

Source: <https://www.simplyrecipes.com/your-guide-to-sweet-potatoes-5208877>

Activity 4.1

1. Use different learning sources, including your experience from your community, library and internet searches, to find out:
 - (a) The main uses of potato;
 - (b) How sweet potato contributes to food security in households and the community; and
 - (c) Different products made from sweet potatoes.
2. Summarise your observations in your portfolio

Principles and practices of sweet potato production

Choosing a site for sweet potato production

Sweet potatoes thrive in well-drained sandy loam or loamy soil with a pH of 5.5 to 6.5. Heavy clay soils are not recommended because they retain water, leading to root rot. They require about 500 to 900 mm of rainfall during their growing season. Sweet potatoes require warm climates with temperatures ranging from 24-35°C.

Selection of planting material

Vegetative propagation is the common method used to propagate sweet potato in Tanzania. Young vines (apical cuttings) are recommended because cuttings from the base of the vines frequently harbour weevils. Disease-free plant materials can be produced in the laboratories using tissue culture technique. Using clean, disease-free planting materials from reliable sources is essential to increase sweet potato production. The vines of length (15-50 cm) are prepared and used for establishing a field for sweet potato (Figure 4.2).



Figure 4.2: sweet potato vines

Source: <https://www.johnnyseeds.com/growers-library/vegetables/sweet-potatoes/growing-tips-sweet-potatoes-from-slips.html>

Land preparation

Land preparation for planting sweet potato is prepared in various ways. It is usually prepared when the soil is moist and workable at the beginning of rains, though it can be carried out at any time. Four seedbed types are used in potato production: ridges, mounds, raised, and flatbeds. Ridging is the most common method of land preparation in Tanzania.

Farmers prepare ridges by scraping weeds into the old furrows and then splitting the old ridges to form new furrows, covering the weeds. Ridges protect the soil and improve drainage. Mounds are made by hoeing the soil together and their surrounding area. They are often found in intercropped fields with cassava or other crops. The following are steps to be followed to prepare a ridge for planting sweet potato:

- (i) Choose a well-drained area with sandy loam or loamy soil;
- (ii) Remove any vegetation, weeds, and debris from the site to ensure a clean working area;
- (iii) Plough the land to a depth of 20-25 cm to break up the soil and improve aeration; then,
- (iv) Form ridges approximately 20-25 cm high and 30-40 cm wide spaced about 90-120 cm apart to allow for vine growth and ease of access.

Planting sweet potato

It is advisable to plant sweet potato vines in the morning/evening, although it can be done in the afternoon, depending on the weather.

Procedure for planting sweet potatoes

- (i) Collect cuttings from healthy mother potato vines;
- (ii) Insert cuttings into the soil to a depth of 7.5-10 cm and 30-40 cm spacing between plants in a ridge (Figure 4.3);
- (iii) Firm the soil around each cutting to ensure good contact; then,
- (iv) Keep the soil moist but not waterlogged, especially during the first few weeks after planting.

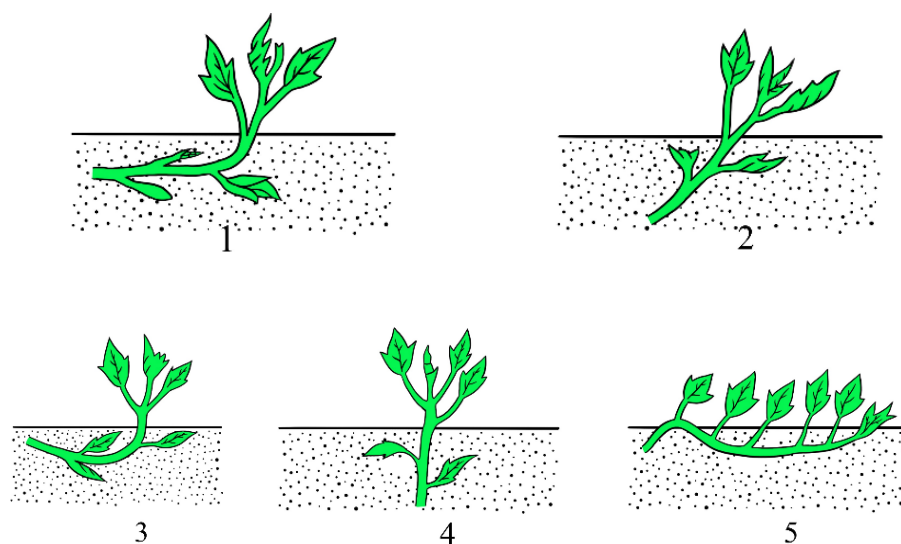


Figure 4.3: Different ways of planting sweet potato

Soil water management

Maintaining consistent soil moisture after planting is vital to establish strong roots and vines. Supplemental irrigation may be needed in areas with insufficient rainfall to maintain soil moisture levels. A farmer can adopt irrigation techniques such as furrow, sprinkler, or drip irrigation. As the crop matures, irrigation should be reduced to enhance tuber development and minimise disease risks. Mulching using organic mulch (e.g., straw, grass clippings) is recommended to conserve moisture and regulate temperature in the soil.

Activity 4.2

In your school field, perform the following tasks:

- (a) Select a suitable area and prepare the land for planting sweet potato;
- (b) Identify the best variety of sweet potato of your choice;
- (c) Prepare the holes at the recommended spacing;
- (d) Plant the selected vines; and then,
- (e) Manage the plant and keep a record in your portfolio.

Exercise 4.1

1. What is the expected sweet potato yield in a one-hectare field?
2. Outline tools and implements used in land preparation for the sweet potato.
3. Describe the steps involved in preparing ridges, mounds, raised and flatbeds for planting sweet potatoes.

Nutrient management

Nutrient management in sweet potato cultivation is essential for achieving optimal growth, yield, and quality. Therefore, nutrients should be applied to sweet potatoes at the right time and in the right amounts. It is recommended to incorporate organic matter, such as compost, and farm yard manure or crop residues during land preparation to improve soil structure, fertility, and moisture retention. Sweet potato requires macronutrients and micronutrients. The macronutrients include nitrogen (N), phosphorus (P), potassium (K), sulfur (S), calcium (Ca), and magnesium (Mg). While micronutrients include manganese (Mn), zinc (Zn), iron (Fe), copper (Cu), molybdenum (Mo), and boron (B). Effective nutrient management in sweet potatoes involves soil testing, appropriate fertilisation based on the fertility status, proper timing, and irrigation practices. For more recommendations, see the agricultural field workers.

Pest management

Various pest species can infect and compromise sweet potato plant growth and productivity. Sweet potato diseases may be caused by three general groups of pathogens: fungi, nematodes, and viruses, while bacterial diseases are minor. Weeds, insects, and vermin are the most common pests that affect sweet potatoes. The management of these pests is discussed here.

Weed management

It is advisable to keep the area weed-free, especially in the early stages of growth. Weeding frequencies range from 1-3 times. The first weeding is done between 1-2 months after planting. During this time, hand weeding or shallow cultivation is recommended to avoid damaging roots. Weeds can also be suppressed in sweet potato farms by mulching (using organic mulch, e.g., straw, grass, and clippings).

Management of other pests and disease

Various soil insects, including sweet potato weevils, rootworms, wireworms, white grubs, fringed beetles, and flea beetles, can damage the sweet potato root. Only the sweet potato weevil larvae tunnel throughout the root, while other soil insects feed on the surface of the developing root.

(a) The sweet potato weevil: The sweet potato weevil (*Cylas formicarius*) is a tiny insect, about 6-8 mm long, with a distinctively elongated body and a dark metallic blue or black colour (Figure 4.4 (a))



Figure 4.4 (a): Adult sweet potato weevil



Figure: 4.4 (b): Larva of sweet potato weevil



Figure 4.4 (c): Damaged sweet potato tubers

Source: https://ucnfanews.ucanr.edu/Articles/Insect_Hot_Topics/http___ucnfanews.ucanr.edu_Articles_Insect_Hot_Topics_INSECT_HOT_TOPICS_Sweet_potato_and_rough_sweet_potato_weevils/

Management of sweet potato weevil: This pest can be managed by planting weevil-free planting materials and avoiding rotating sweet potatoes with alternative host crops to disrupt the weevil life cycle. It is recommended that plant residues be removed and destroyed after harvest to eliminate potential breeding sites. Ensure that tubers are well-covered with soil, and use mulch to prevent weevils from accessing the tubers. Use pheromone traps to monitor and reduce adult weevil populations. Harvest sweet potatoes when they reach maturity to reduce their exposure to weevils. Store harvested sweet potatoes in clean, ventilated, and weevil-free environments.

(b) White grub

White grubs are the larvae of various beetle species, such as scarab beetles (Figure 4.5). They feed on sweet potato roots, causing significant crop damage by disrupting root systems, which can lead to reduced plant growth and yield.



Figure 4.5: White grub

Source: <https://www.gardentech.com/insects/white-grubs>

Management of white grub: This can be managed through crop rotation and soil cultivation to expose the larvae to predators and use natural enemies such as starling birds and ground beetles.

(c) Flea beetles

Fleas are small jumping beetles about 1-3 mm long with shiny black, brown, or metallic-coloured bodies (Figure 4.6). They feed on the leaves of potato plants, causing small holes and stippling, which can weaken plants and reduce yields. Their larvae damage sweet potato tubers.

Management of flea beetles: Effective management includes using row covers, rotating crops, and applying insecticides if necessary. Consult agricultural experts for recommended insecticide to use against flea beetles in potatoes.



Figure 4.6: (a) Adult flea beetle



Figure 4.6 (b) sweet potato affected with a larva of flea beetle

Sources: <https://extension.usu.edu/vegetableguide/potato/flea-beetles>

Sources: <https://www.ipmimages.org/browse/detail.cfm?imgnum=5605714>

Viral diseases

Sweet potatoes are affected by viral diseases such as *Sweet potato mild mottle virus* (SPMMV) (Figure 4.7 (a)) and *Sweet potato feathery mottle virus* (SPFMV) (Figure 4.7 (b)). These diseases are mainly transmitted from infected to uninfected plants through vectors and planting materials.



Figure 4.7 (a): Sweet potato mild mottle virus symptoms on leaves



Figure 4.7 (b): Sweet potato feathery mottle virus symptoms on leaves

Source: <https://www.cabidigitallibrary.org/doi/abs/10.1079/cabicompendium.51028>

Management of viral diseases: These diseases can be managed by planting virus-free sweet potato cuttings obtained from reputable sources; rotating sweet potatoes with non-host crops to disrupt the virus transmission cycle; removing and destroying infected plants and crop residues to minimise sources of viral infection; controlling weeds that could serve as alternative hosts for viruses and vectors.

Activity 4.3

1. Pay a visit to your school farm or nearby sweet potato fields; observe and perform the following tasks:
 - (a) Identify some common pests and diseases affecting a sweet potato crop;
 - (b) Observe effects caused by the identified pests;
 - (c) Study methods that are used to manage the identified pests, including vermin; and,
 - (d) Adopt manageable measures and apply them in the field for a sweet potato.

Other agronomic practices

Intercropping is another managerial practice commonly applied in sweet potato fields. The crop can be intercropped with cassava, maize, beans, cowpeas, and groundnuts (Figure 4.8). This practice reduces pest and disease incidence and improves soil fertility and productivity.



Figure 4.8: Sweet potato intercropped with maize

Source: <https://farmbizafrica.com/maize-sweet-potato-intercropping-improves-income-nutrition/>

Harvesting

Potatoes can be harvested by hand or machine. The leaves of mature potatoes turn yellow, typically around 90-120 days after planting. In small-scale production, sweet potatoes can be harvested by carefully digging up the tubers using a hand hoe, spade, or fork to avoid bruising (Figure 4.9 (a)). However, using a harvester (Figure 4.9 (b)) is more convenient on a large scale.



Figure 4.9 (a): Harvesting using a hand tool

Source: <https://www.bhg.com/how-and-when-to-harvest-sweet-potatoes-7642613>



Figure 4.9 (b):Harvesting using sweet potato harvester

Source: <https://www.stricklandbros.com/short-bed-digger.html>

Post-harvest management

Sweet potato tubers are gently cleaned to remove excess soil without washing to reduce the incidence of moisture-related decay. Cure the potatoes in a warm, humid environment (26-29°C with 85-90 % humidity) for 1-2 weeks to enhance sweetness and extend storage life. Tubers are then graded, ensuring any damage or disease is discarded. Finally, cured sweet potatoes should be stored in a cool, dry place (around 13-16°C) with good ventilation, avoiding direct sunlight and excessive moisture.

Activity 4.4

1. Visit your community and learn about:
 - (a) Methods of processing sweet potato;
 - (b) Categorise the methods into traditional and modern ones; and,
 - (c) Select one method of your choice and use it to process sweet potato.
2. Summarise what you have learnt from this activity in your portfolio.

Exercise 4.2

1. Identify other diseases affecting sweet potato in the crop field besides viral diseases.
2. Outline key challenges affecting sweet potatoes in the field.
3. Explain how to manage water stress in the sweet potato field.

Marketing

Marketing sweet potatoes can involve several strategies to ensure the produce and its products reach consumers at optimal conditions and generate profit. The strategies include:

- (i) Identifying target markets by determining potential buyers such as local markets, restaurants, and food processors.
- (ii) Sorting and grading sweet potatoes based on size, shape, and quality to meet market standards.
- (iii) Analysing sweet potato demand by understanding consumer preferences, market trends, and seasonal demand fluctuations.

- (iv) Using appropriate packaging materials to protect sweet potatoes during transportation and storage; labelling the packages with relevant information such as variety, grade, and weight.
- (v) Developing a strong brand identity to distinguish your sweet potatoes from competitors.
- (vi) Using appropriate distribution channels such as direct selling to consumers, wholesalers, and food processors in bulk quantity.
- (vii) Establishing reliable distribution networks to ensure timely delivery to various markets.

Activity 4.5

1. Visit a large-scale sweet potato producer and gather detailed information on all key aspects of production. Then, based on the insights gained from this study visit, develop a project plan that can be used to start a similar project of your own.
2. Use the records you kept in various aspects of sweet potato production to perform the following tasks:
 - (a) Evaluate the profitability of sweet potato production;
 - (b) Prepare a report showing the challenges and successes of sweet potato production; then,
 - (c) Present the report in class.



In your portfolio, summarise what you have learnt from this chapter.

Chapter Five

Round potato production

Introduction

In this chapter, you will learn the principles and practices of producing round potatoes. The competencies developed will enable you to produce, process, store, market, and keep records when producing round potatoes. You will also explore various opportunities in the value chain of potato production.



Think

“Popular chicken chips preferred by most young people.”

Round potato and its importance

The round potato (*Solanum tuberosum* L.), also known as the Irish potato, is an annual crop belonging to the Solanaceae family. The crop is grown for its tuber (an underground stem) (refer Figure 5.1 (b)). In Tanzania, round potato is cultivated in the Southern Highlands of Tanzania (Mbeya, Iringa, and Njombe regions). Other regions are Kagera, Arusha, and Tanga (Usambara mountains). Round potatoes are rich in starch, protein, vitamin C, calcium, phosphorus, and trace elements (boron, manganese, copper, iron, and zinc).

Due to an increasing population globally, the demand for round potato products is expanding rapidly. In Tanzania, round potatoes are essential for food security and improve nutrition, employment, and income.

Classification of round potato varieties

Round potatoes are classified according to their uses. Table varieties are mainly for cooking, mashing, roasting, baking, and stewing. They are generally cultivated

for the fresh market. Table varieties include Asante, Meru, Sherehekea, and Saggita. Chips varieties such as Tengeru, Rumba, and Panamera are the most appropriate for making chips; Markies Varieties are used for making thin, round fried potato slices (crisps).



Figure 5.1(a) A field of round potato

Source: <https://spudsmart.com/strategies-for-protecting-potato-crops-from-key-weeds/>



Figure 5.1(b) Round potato tubers

Source: <https://www.apnikheti.com/en/pn/agriculture/horticulture/vegetable-crops/potato>

Activity 5.1

Use various learning resources, including an atlas and internet searches, to perform the following:

- Identify regions with high production of round potatoes.
- Suggest measures that can be taken to improve the production of round potatoes in Tanzania.
- Summarise what you have learnt in your portfolio.

Principles and practices of growing round potato

Adhering to principles and good agronomic practices (GAPs) increases its productivity. The following sub-sections present GAPs underlying the production of round potatoes.

Choosing a site for round potato production

The crop grows in well-drained, loamy soils rich in organic matter and requires an annual rainfall of 800 -1750 mm. In Tanzania, round potatoes grow well at altitudes ranging from 1200 to 2700 m above sea level. The optimal temperature for growing round potatoes is 15 - 20 °C and soil pH ranges from 5.5 to 7.0. In soils with a pH value less than 5.0, limestone should be applied at 1 ton per hectare to improve round potato production.

Activity 5.2

Through the library, internet searches, and other learning resources:

1. Find out information concerning the agroecological conditions of major areas where round potato are produced in Tanzania;
2. Assess the environmental conditions of your location and determine its suitability for the production of round potato; then,
3. Summarise your observations in your portfolio.

Land preparation

The field for round potatoes should be cleaned well. After clearing, the next step is tilling; this helps to mix the soil, reduce weeds, improve water infiltration and soil aeration, and mix the remains of the previous season's crops with the soil. Till the soil to a 30-45 cm depth to create a fine seedbed. Avoid burning residues in the field to protect the environment and the organisms contributing to soil fertility. Round potatoes have a higher demand for nitrogen (N), phosphorus (P), and potassium (K) than for other plant nutrients. Improve the soil structure by adding organic matter or compost. In areas where manure is available, 25 tons of well-rotted manure per hectare will improve the production of round potatoes and other crops that will follow in the cropping cycle. It is essential to apply fertilisers based on the soil test results of a given field.

Selection of planting materials

Choose round potato varieties suited to local conditions, market demand, and the intended use, such as processing or fresh market. Purchase certified seed potatoes from reputable dealers to maximise round potato production by minimising the risk of introducing diseases. The improved varieties registered by the Tanzania

Official Seed Certification Agency (TOSCI) include Sagitta, Sherekea, Taurus, Challenger, Rumba, Rodeo, Panamera, Markies, Manitou, Jelly and Asante. Seek advice from reliable sources (Agricultural experts, Research Institutions, nearby farmers, seed sellers, parents, and your teacher) for promising round potato varieties. Good quality seed potato is of known variety (not mixed with other types), resistant to diseases, uniformly sized 35-45 mm (chicken egg size), and well sprouted. Figure 5.2 represents good quality round potato planting materials.



Figure 5.2: Good quality seed potatoes

Activity 5.3

1. Use various learning materials, including the internet, to prepare a summary that can be used to advise farmers on selecting good-quality seed potatoes.
2. Visit one of the seed potatoes companies or a farmer involved in round potato seed production, study the procedures applied to produce quality seeds, and share with your fellow students.
3. Identify which round potato varieties are much more widespread in Tanzania.
4. Find out why the identified potato varieties in 3 above are popular in the country.

Planting

The appropriate time for planting round potato vary based on altitude and rainfall patterns. In the Southern Highlands, potatoes are typically planted in September or October at higher altitudes (above 2000 meters a.s.l.). In the mid-latitude regions (1500-2000 meters a.s.l.), planting begins after the start of the rains in November or December, allowing for a second planting in February. In the Northern Highlands, potatoes are usually planted when moisture conditions are favourable. The rainfall in this region is divided into two seasons, enabling

multiple crops per year. Planting during the long rainy season (Masika) typically begins with the onset of rains in March or April, while planting during the short rainy season (Vuli) is done in November or December. The recommended spacing for a round potato is 70 cm between rows/ridges and 30 cm between plants (70 cm × 30 cm).

Procedure for planting

- (i) Open furrows 10-15cm deep at the spacing of 70 cm between furrows;
- (ii) Apply the recommended planting fertiliser in furrows;
- (iii) Cover the fertiliser with a thin layer of soil;
- (iv) Place a selected seed potato in a furrow with sprouting buds up at a spacing of 30 cm between plants;
- (v) Cover the seed potato with the remaining soil; then,
- (vi) Irrigate the field if the soil is not wet.

Abiding with the recommended spacing is essential to ensure sufficient space for growth and simplify weeding, fertiliser, and pesticide application. For optimal production, the ridge method of planting is recommended. For a 35-45 mm seed size, you will need 2000-2500 kg of seed potatoes per hectare. High yields of round potatoes are achieved through effective agricultural practices, such as using quality, disease-free seeds that are 3 to 5 months old. Figure 5.3 shows seed round potatoes placed in furrows.



Figure 5.3: Seed round potatoes placed in furrows

Activity 5.4

1. Visit your school farm and then do the following.
 - (a) Using the criteria, you have learned in the class, select a suitable area for round potato establishment;
 - (b) Prepare the land accordingly; then,
 - (c) Select the best round potato variety and follow the procedures to plant them in your established field.

Exercise 5.1

1. Based on your knowledge, how would you advise a farmer who seems to be against certified seed potatoes?
2. Using a spacing of 70 cm x 30 cm. Calculate the plant population in one hectare of round potato field.
3. Explain why round potato seeds can sprout even without being planted in the field.
4. Differentiate between potato seeds and seed potatoes.

Soil water management

Round potatoes need consistent moisture, especially during tuber formation. Avoid waterlogging, which can lead to diseases. The use of drip, sprinkler, or furrow irrigation to supply moisture in the field can enhance water use efficiency. Low and variable soil moisture levels lead to reduced dry matter and poor tuber set. Planting potatoes on mounds also enhances water-soil conservation.

Activity 5.5

1. Visit some nearby small and large-scale potato fields or use internet/library resources to study soil water management practices applied to round potato crop plants.
2. Design an appropriate soil water management strategy and apply it to the crop plants you have established on the school farm.

Nutrient management

Organic and inorganic fertilisers are used to improve yield in round potato production. Their application rates will vary depending on soil test results specific to an area. Organic fertilisers include farm yards and compost manures, which are slow-release organic fertilisers that can be used. Compost is a decomposed mixture of organic residues (manure, straw, ashes, kitchen wastes). Fresh manure can burn seed round potatoes when it comes into contact with them or if placed too close. Various types of industrial fertilisers are used at different stages of plant growth. For instance, diammonium phosphate (DAP) is applied during the crop planting. At the same time, fertilisers like calcium ammonium nitrate (CAN) and urea are used to top-dress later in the growth cycle. When growing round potatoes, it is recommended to apply DAP fertiliser as advised by an agricultural expert at a rate of 250-375 kg per hectare. Apply fertiliser per planting hole on the furrow and cover it lightly before planting tubers to prevent fertiliser-seed contact that may burn seed potatoes. Avoid using excess nitrogen, as it promotes vegetative growth at the expense of tuber formation.

Alternatively, use either DAP or a combination of Triple Superphosphate (TSP) with Calcium Ammonium Nitrate (CAN) per hectare in the furrows during planting as per recommendation after testing the soil. If the soil is low in potassium, farmers should use Muriate of Potash. Root crops require more Potassium (K) than Nitrogen (N) and Phosphorus (P), so it's essential to apply the recommended amount of Potassium per hectare. Another suitable fertiliser is NPK. However, the soil fertility status should determine the type and amount of fertiliser to be used. You should consult an agricultural expert in your area for the appropriate variety and fertiliser rate.

Activity 5.6

1. Through farm visits, internet searches, library, and market surveys:
 - (a) Identify different types of organic and inorganic fertilisers that are recommended for round potato production;
 - (b) Estimate the amount and the right time of application for each fertiliser to potato plants;
2. Under the guidance of your teacher, choose and apply the right fertiliser in your plot accordingly; and,
3. Keep on observing and recording the development of the crop plants.

Pest management

Round potatoes are affected by various pests, both in the field and during storage. The most common pests affecting potatoes are weeds, pathogens (fungal and bacterial), nematodes, and insect pests. These pests can cause significant destruction of the crop. Therefore, it is essential to provide appropriate management based on the crop in question.

Weed management

Keep the round potato field weed-free. Weeds should be removed to avoid competition with the crop for water, light, nutrients, and space, leading to reduced yields. Weeding should start 3-4 weeks after planting (1-2 weeks after germination), depending on the weather or when weeds appear. A hand hoe is recommended for weeding as it can remove all weeds. This weeding should be done before the round potatoes start to set to avoid damage to small potato tubers.

Additionally, non-selective herbicides can be used to kill perennial weeds before land preparation or before planting. Use another type of herbicide immediately after transplanting. If possible, cover the soil with mulch to suppress weed growth. Combine different methods to control weeds instead of relying on a single method. You can seek advice from an agricultural extension worker on the type of herbicide to use against the available weeds on your farm.

Activity 5.7

Visit the school field where the round potato has been planted and do the following:

- (a) Identify the types of weeds in your school's field for the round potato;
- (b) Recommend methods of weeding depending on the weeds found on the farm;
- (c) Under the guidance of your teacher, apply a selective herbicide to one section of the field while using hand hoe weeding in another section; then,
- (d) Compare the effectiveness of the two methods.

Insect pest management

Effective insect pest management involves strategies to control insect pests that can harm crop yield and quality. Whiteflies and potato moths are among the important insect pests in the round potato crop. They transmit pathogens causing

viral diseases. Fundamental approaches for controlling the round potato insect pest include biological, cultural, mechanical, and chemical methods. Insect pest management is described in the following subsections.

(a) Whiteflies

Whiteflies (Figure 5.4) are common pests in potato crops. They feed on plant sap, excrete sticky honeydew, and can transmit plant diseases.



Figure 5.4: White flies underneath round potato leaves

Management of whiteflies: Inspect the crop plants regularly for whiteflies and their eggs, especially the undersides of leaves, and apply appropriate management measures. Avoid planting potatoes near crops that are known to attract whiteflies, such as tomatoes or peppers. Use yellow sticky traps to capture adult whiteflies. Apply insecticidal soaps or neem oil, which are less harmful to beneficial insects. Use systemic insecticides as a last resort, following the recommended guidelines and safety precautions. Combine multiple control methods for effective and sustainable whitefly management.

(b) Potato moth

Potato moths (Figure 5.5 (b)) attack potatoes from the field and in the warehouse. They undergo complete metamorphosis (egg-larvae-pupa-adult) and are more common in warm, dry, and hilly areas. The significant damage caused by the moth is making holes in the potatoes that pave the way to fungal and bacterial diseases, leading to potatoes rotting (Figure 5.5 (a)).



Figure 5.5 (a): potato tubers damaged by moth larvae



Figure 5.5 (b) adult moth

Management: This pest can be managed through crop rotation, deep planting (10 cm), compact hilling, destruction of infested plant material, proper storage, and use of recommended insecticides. Healthy plants and clean seeds are recommended because infested seed tubers are the leading cause of re-infestation in the field.

Diseases management

Round potato is susceptible to various diseases that result in reduced yield and lower quality of the potato tubers. Early and late blight potato diseases, bacterial wilt, and viral diseases are the most harmful. Round potato diseases can be controlled by using resistant planting materials, crop rotation, removing residues in the field, and integrating different management practices.

(a) Late blight

Late blight (Figure 5.6) is one of the most destructive fungal diseases of round potatoes caused by the pathogen *Phytophthora infestans*. Cool, moist conditions favour this fungus and can lead to significant yield losses if not managed properly.

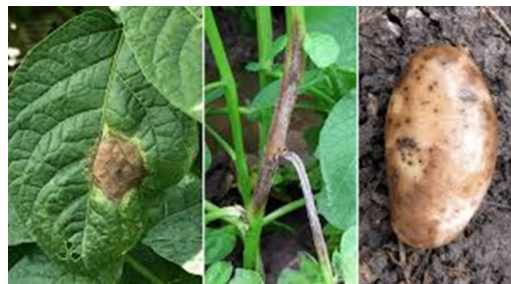


Figure 5.6: Late blight in potato leaf, stem, and tuber

Symptoms of the disease: Dark, water-soaked lesions appear on leaves, beginning at the tips and rapidly expanding, turning brown and causing leaf death. Stems develop dark brown to black lesions, which can girdle and kill the plant. Infected tubers become firm brown and develop corky spots beneath the skin, which may become more extensive and lead to rot and secondary infections.

Management of the disease: Apply the recommended fungicides as a preventive measure or at the first sign of disease. Adopt improved cultural practices such as crop rotation, field sanitation, plant spacing, and irrigation. Use round potato varieties that are tolerant to late blight. Harvest tubers when vines are dry to reduce the risk of tuber infection. Store potatoes in cool, dry conditions and regularly inspect stored potatoes for signs of late blight. Clean the working tools and equipment to prevent the spread of the pathogen. Combining these strategies allows you to effectively manage late blight and minimise its impact on your crops.

(b) Early blight potato disease

Early blight potato disease is a fungal disease caused by the pathogen *Alternaria solani*. It is favoured by warm temperatures (24-29°C) and high relative humidity.



Figure 5.7: Early blight in potato plant and tuber

Symptoms of the disease: Dark, concentric rings appear on infected leaves. These spots usually start on older leaves and then progress to younger foliage. Dark, sunken spots may also strap the stem, leading to wilting and death of the plant (Figure 5.7).

Management of the disease: Monitor and look for the early signs of disease to initiate early control measures. Ensure adequate fertilisation, particularly with potassium and phosphorus, to maintain plant health and reduce susceptibility. Apply the recommended fungicides as a preventive measure or at the first sign of disease. Adopt improved cultural practices such as crop rotation, field sanitation, plant spacing, and irrigation. Use round potato varieties that are tolerant to early blight. Effective management of early blight involves an integrated approach combining cultural practices, resistant varieties, and timely chemical applications.

(c) Bacterial wilt

Bacterial wilt of potato is a severe disease caused by a soil-borne bacterium called *Ralstonia solanacearum*. The disease spreads through contaminated soil, water, tools, nematodes, and plant materials.



Figure 5.8: Bacterial wilt disease in potato plant and tuber

Symptoms: Initial signs include wilting of the leaves during the hottest part of the day, which may recover at night. As the disease progresses, wilting becomes permanent. A brown discolouration can be seen in the vascular tissues of the stem when cut open. When infected with the bacteria, a white, milky, sticky ooze may be visible from cut stems or tubers (Figure 5.8).

Management: The disease can be managed through crop rotation, field sanitation, use of resistant varieties, eradication of infected plants, and proper nutrient and water management.

Viral diseases of round potato

Like other crops, round potatoes are susceptible to viral diseases that can significantly impact yield and quality.

Symptoms of the diseases: Symptoms include mottling, mosaic patterns on leaves, upward rolling of leaves, chlorosis (yellowing), stunted growth, leaf drop, and reduced tuber yield.

Management of viral diseases: Use virus-free certified seed potatoes to reduce the risk of introducing viral pathogens into the field. Control insect vectors like aphids through integrated pest management (IPM) strategies, such as insecticides, natural predators, and cultural practices. Rotate potatoes with non-host crops to break the life cycle of soil-borne pathogens and vectors. Regularly clean and disinfect tools, machinery, and storage facilities to prevent the mechanical spread of viruses.

Nematode diseases management

Potato nematodes seriously threaten potato crops, causing significant yield losses if not managed properly. The main types of diseases caused by nematodes are cyst nematodes (*Globodera* spp.) (Figure 5.9) and root-knot nematodes (*Meloidogyne* spp.). Potato cyst nematodes cause stunted growth, yellowing leaves, poor root development, reduced yield, and form cysts on roots. Meanwhile, root-knot nematodes cause root galls, stunted growth, wilting, and deformed tubers.

Management of the diseases: To control nematode diseases, use resistant varieties, crop rotation, biological controls, and soil fumigation. Key preventive measures include soil testing, using nematode-free seed potatoes, and maintaining field hygiene. An integrated management approach is essential for effective nematode control and minimising crop losses.



Figure 5.9 (a): Root-knot nematode disease

Source: <https://veggiescout.ca.uky.edu/root-knot-nematodes-potato>



Figure 5.9 (b): Nematode cysts on potato roots

Source: <https://www.science.org/content/article/potato-farmers-conquer-devastating-worm-paper-made-bananas>

Activity 5.8

Visit the round potato field you have established and do the following practices:

- (a) Identify the pests available in the field at your school;
- (b) Identify the suitable management measures for the identified pests; and
- (c) Apply the identified management measure in the established field and evaluate its effectiveness.

Harvesting

The days to maturity on round potatoes depend on the variety. Early-maturing round potato varieties will be ready in 3 months, medium-maturing varieties will take 3 to 4 months, and late-maturing round potato varieties will take more than four months to mature from planting.

Potatoes are ready for harvesting when the leaves are dry, have turned yellow, and when the potato's skin does not come off quickly when rubbed with the fingers. Additionally, the potatoes should easily separate from the roots. Harvesting can be done by hand or machine (Figure 5.10 (a) and (b)).



Figure 5.10 (a): Potato harvesting by hands **Figure 5.10 (b):** Mechanised potato harvesting

Source: <https://hzpt.com/potatoharvester/>

Immaturely harvested tubers usually have lower dry matter content. They also tend to experience more skin damage, making them more susceptible to infections by fungal and bacterial pathogens.

Cutting or detaching the vegetative part of potato plant above the ground (dehaulming) two weeks before harvesting helps to harden the tuber skins. Hardened skins reduce the risk of damage during harvesting and post-harvest handling. Mature potatoes can be dehaulmed and left in the soil for 1–2 months. The average yield of round potatoes in Tanzania is 8.43 t/ha for small-scale farmers. Proper farming and uses of clean planting materials can increase yields to 15 – 20 t/ha.

Postharvest management of round potato

After harvesting, some processes need to be done on the produce to improve its shelf life. These include curing, sorting and grading, packaging, and marketing.

- (a) **Curing:** This is the process of enhancing the shelf life and quality of harvested potato tubers by spreading them under shade to allow healing of minor injuries and hardening their skins. Depending on weather conditions, curing can be done in the field or a specific place.
- (b) **Sorting and grading:** After round potato tubers have been cured, they should be sorted and graded based on their size, shape, and quality, ready for market (refer to Figure 5.11).



Figure 5.11: Sorting and grading of round potato tubers

(c) **Packaging:** Net bag packaging (nylon mesh bags) is recommended for packing round potato tubers because they provide good ventilation (Figure 5.12)



Figure 5.12: Potato packaging net

(d) **Marketing**

Effectively marketing potatoes requires a combination of strategies. Here are some of the strategies:

- (i) Understand the potato market: Conduct surveys to understand consumer preferences, buying habits, and trends.
- (ii) Product differentiation: Promote different varieties of round potatoes and their specific uses. It is important to offer convenient packaging sizes for various consumer needs.
- (iii) Branding and messaging: Convince the market by creating a story about the potatoes and highlighting, e.g., health benefits and organic certification. Showcase how potatoes can be prepared (baked, mashed, fried, or roasted).
- (iv) Marketing channels: Use platforms like Instagram, Facebook, and WhatsApp to reach a large audience.
- (v) Partnerships and collaborations: You can collaborate with restaurants and catering service providers to feature potatoes in their dishes.
- (vi) Promotions and events: Offer discounts and participate in farmers' markets to engage directly with consumers.
- (vii) Advertising: Create and use online food blogs and social media accounts. Also, posters can be fixed strategically to reach a broader audience.

Activity 5.9

Visit the nearby market and perform the following tasks:

- (a) Identify the products of round potato present in the market and compare their prices;
- (b) Read the labels of different potato products packaging materials and summarise the information provided;
- (c) Conclude the best potato product that can give higher benefit compared to others; then,
- (d) Summarise your observations and share in class.



In your portfolio, summarise what you have learnt from this chapter.

Chapter Six

General principles and practices of livestock production

Introduction

This chapter will introduce you to the principles and practices of raising and keeping livestock. The competencies developed from this chapter will enable you to apply the principles and practices in raising any livestock for commercial and domestic purposes.



Think

“A livestock keeper with just ten improved cows per herd can produce more milk than one with twenty unimproved cows per herd.”

Concept of the principles and practices of livestock production

Livestock refers to the domesticated animals raised in an agricultural setting in order to provide labour and diversified products for economic purposes (Figure 6.1). Livestock production is the science and art of keeping livestock. As a science, livestock production follows a particular set of rules which, when followed accordingly, can lead to the best performance in livestock production.



Figure 6.1: Some classes of livestock

Activity 6.1

Visit a school farm or any nearby livestock farm or use a library/internet search to:

1. Inquire/study a series of management practices that are taking place on the farm.
2. Summarise the observed practices in your portfolio.

The principles and practices of livestock production

For a farmer to raise any livestock enterprise successfully, there are principles and good practices to follow and abide by. The principles include:

- (a) Planning for livestock production;
- (b) Choosing the suitable breed;
- (c) Livestock housing;
- (d) Feeding management;
- (e) Disease management;
- (f) Reproduction and breeding of livestock;
- (g) Processing and marketing of livestock products; and
- (h) Livestock farm record keeping.

Planning for livestock production enterprise

Livestock enterprise means producing livestock products by using and managing land, labour, capital, and other essential production inputs. Since these production resources are scarce, planning is an essential and foremost activity for any livestock enterprise.

Planning requires an individual to have the vision or ability to see the enterprise's success before implementation. Knowing where the enterprise will be and what to achieve in three to five years. Planning entails considering, among others, why you want to begin a livestock enterprise, the available resources to support the enterprise, the local markets you have access to, and the risk you will bear.

Current experience in climate change makes planning even more crucial. Like all other countries, Tanzania is currently experiencing severe weather extremes due to a changing climate. The increased temperatures, prolonged droughts, floods, and erratic rainfall affect livestock feed resources and significantly impact livestock

production. The planning process needs to take this into consideration and set control measures to reduce the negative impacts. The measures can include:

- (a) Using drought-tolerant forage varieties;
- (b) Implementing water management strategies (irrigation and water conservation measures);
- (c) Implementing forage conservation measures; and
- (d) Selecting more tolerable animal breeds.

In addition, farm record-keeping should be taken into consideration at each stage of crop production. Figure 6.2 summarises some factors to consider when planning for a livestock enterprise.

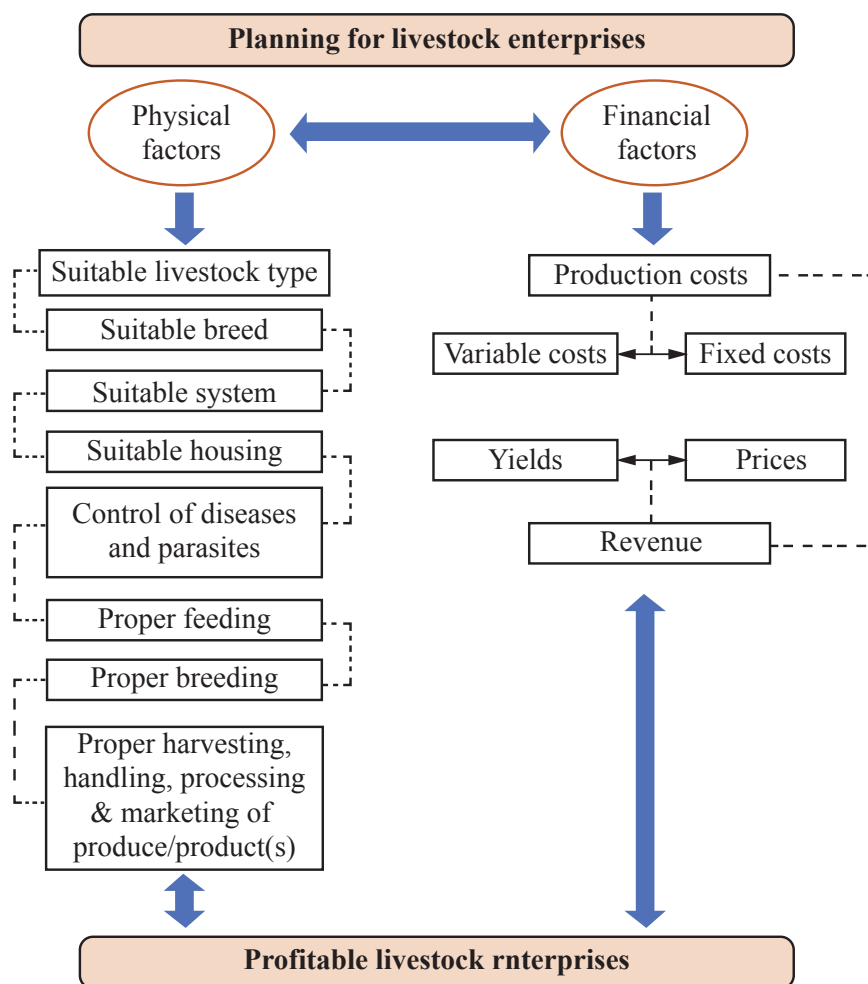


Figure 6.2: Factors to consider when planning for a livestock enterprise

Choosing the suitable breed

Selection of the breed is a personal preference based on objectives, favourable livestock-keeping conditions and interests. This is extremely important to consider as every animal has different requirements, so doing a thorough study on the animal of your choice is recommended. For example, if you want to raise dairy cattle breeds, you will need a different type of house and management than if you were raising beef cattle breeds. You'll also need to purchase various kinds of feed and bedding material. So, think exhaustively about what type of livestock to be raised before making any decisions. All traits of economic value, should be considered when selecting any breed of animal. The major traits influencing productive efficiency in animals are reproductive performance or fertility, maternal ability, milk production, meat deposition, carcass quality, egg production, egg quality, wool production and growth rate.

Livestock housing

Livestock housing refers to confining an animal in a built enclosure. A house for livestock plays a vital role in protecting the livestock from extreme heat, cold, wind, sunlight, and rain. It protects the animals from theft and wild animal attacks. It provides a clean and comfortable environment and facilitates efficient livestock management.

Generally, before constructing any livestock house, you need a farm layout plan to indicate the position of the animal houses/buildings corresponding to their uses. Some critical factors to consider when planning for farm animal housing include topography and drainage, soil type, exposure to the sun, and protection from wind and rain. Others are accessibility, durability, attractiveness, security, water and power supply.

Characteristics of appropriate housing

Each livestock requires its appropriate housing to enhance the comfortability of the animals and facilitate efficiency when carrying out managerial practices. Generally, proper housing of farm animals should have the following features:

- (a) Freedom to exhibit natural behaviour. This is because every farm animal species has unique natural behaviours that differ from others.
- (b) Easy accessibility to adequate food and water as well as environmental requirements for maintenance of health and vigour.
- (c) Freedom of movement to stand, stretch, and lie down.

- (d) Light during the daylight hours.
- (e) Visual and social contact with other animals.
- (f) Roof and walls that protect from bad weather, which neither harms nor distress.
- (g) Rapid identification and treatment of vices, injuries, diseases, and parasites.
- (h) Well-ventilated: To allow enough air movement in and out of the house.

Activity 6.2

1. Pay a visit to a school farm or nearby farms to:
 - (a) Observe different types of livestock houses; and
 - (b) Comment on the appropriateness of each house to the required standard for each kind of livestock.
2. Summarise your observation in your portfolio.

Feed and feeding management

Feeding farm animals is one of the significant daily activities a livestock farmer must do. Farm animals depend on feeds for their body maintenance and various functions, including production, reproduction, growth, and development. Feed is any nutritional material that can be ingested, digested, absorbed, and utilised by the animal body. Feed quality and quantity (balanced ration) enhance its utilisation and performance. Feeds contain nutrients of different types and quantities. Nutrients are chemical compounds released from the feeds by metabolism processes and utilised by the body for various functions. Such nutrients are also known as feed components. The major components of feeds are generally classified into six groups: water, carbohydrates, protein, fats, vitamins, and minerals.

Before buying/preparing feeds for the animals, it is important to know something about the digestive systems of farm animals. This will guide you on the correct feedstuff required by a specific type of livestock. Livestock may be grouped into two major categories based on their digestive systems. These categories are ruminants and non-ruminants.

Ruminants (including cattle, sheep, and goats) have complex stomachs, and their fore-stomach is divided into four sections. They can utilise nutrients from plant-

based feeds by breaking down the carbohydrate materials through microbial actions. However, digestion in the lower gut of the stomach of the ruminant is enzymatic, just like it is for non-ruminants. Likewise, animals with relatively large caecum (pseudo ruminants) such as horses, donkeys, rabbits and guinea pigs are able to feed and utilise both fibrous and concentrate feeds. On the other hand, non-ruminants, which include pigs, and chickens (animals with simple stomachs), can utilise cereal-based feeds through enzymatic digestion.

Types of livestock feeds

Livestock feeds can be broadly classified into two major groups: roughages and concentrates. The categories of these feeds are summarised in Figure 6.3.

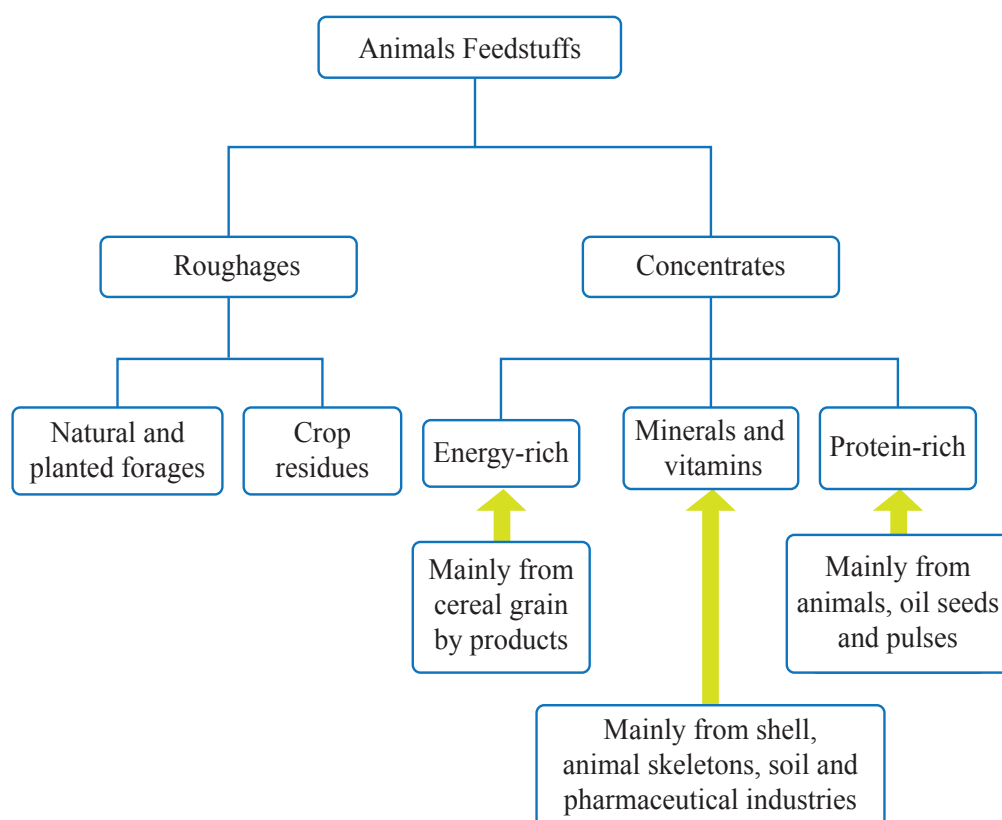


Figure 6.3: Categories of animal feedstuffs

Criteria for feed selection

Livestock farmers must aim to prepare or purchase feeds of good quality, which are cost-effective and meet the requirements of the animal. These feeds must also be liked by the animal. In addition, they can be stored for a desired period. Choosing what feed materials a livestock farmer can use for animals is a decision that must be based on a good understanding of the following criteria:

- (a) The nutritional requirements of the animals;
- (b) The type of animal to be fed;
- (c) Nutritional value of the available feedstuffs;
- (d) Costs of the available feeds;
- (e) Storage ability;
- (f) Availability of feeds;
- (g) Palatability and physical form of the feed; and
- (h) Presence/Absence of anti-nutritional factors.

Livestock feeding systems

Livestock, particularly ruminants, are mainly fed outdoors or indoors. Outdoor feeding involves pasture and rangeland. On the other hand, indoor feeding involves feeding livestock with various feedstuffs in the stalls; hence, it is called stall feeding. The pastures used in grazing ruminants may be natural or established ones. In natural pastures, animals graze extensively and continuously while being provided with water naturally from the rangeland. Sometimes, this system of feeding is called continuous grazing. The established pastures are usually fenced to make paddocks where animals are allowed to feed from one paddock to another in rotation; hence, it is called rotational grazing. In this case, care should be taken to ensure no poisonous plants and that the paddocks are not overgrazed. Water troughs are constructed in the paddocks or at one drinking point. In the stall-feeding system, forages and other feeds are provided in the feeding troughs, while water is provided in the water troughs. The tethering grazing system is sometimes practised in subsistence livestock farming.

Livestock feeding practices

Good livestock feeding practices must ensure proper feed use and maximise productivity. Feeding practices take two forms: free choice (ad-lib feeding), which is practised on young animals and fatteners to promote fast growth and

weight gain, and restricted feeding, which is done on breeders and egg-producing flocks to avoid overweight that can hinder performance. In all cases, restricted feeding is for feed and not water.

Component feeding or total mixed ration can be used to feed ruminants. Component feeding is when only one component of feed ration (foraging), protein supplements, and grains are offered to the animal based on weight. In total mixed ration (tmr), forages, grain and protein by-products (such as sunflower seed cakes maize bran), minerals, and vitamins are mixed to make a balanced ration in which the weight of each ingredient is known.

On the other hand, monogastric feeding practices are either wet-dry feeding or dry feeding. The wet-dry feeding practice uses water to moisten the dry feed to make it easier for the animal to use. The system reduces feed wastage, improves feed efficiency, and increases productivity.

Activity 6.3

Visit a school farm and some nearby livestock farms to study feeding systems and practices on the farm. Then, perform the following tasks:

- (a) Observe the characteristic features of the visited premise;
- (b) Identify the type of livestock feeds and feeding practices employed in the farm; and then,
- (c) Summarize your observation in your portfolio.

Management of livestock health

For an animal to produce optimally, it must be in good health. Health in livestock production implies a state of well-being and normal functioning of body organs and systems. It is freedom from diseases and abnormalities.

Signs of healthy and unhealthy animal

Farm animals need a close follow-up to keep individual animals, flocks, or herds in good health and performing. When the animal falls sick, it shows some signs/behavioural deviation from the normal healthy animals.

Activity 6.4

- Visit a school or nearby livestock farms and do a library/internet search. Then perform the following tasks:
 - Identify signs/behaviours of healthy and unhealthy animals;
 - Identify common vices of the livestock you visited; and
 - Observe how ill health and vices are controlled on the farm.
- Tabulate your observations in task 1 above, as shown here after,
 - Signs/behaviours of healthy and unhealthy animals

Type of livestock	Healthy parameters (where applicable)	Healthy signs/behaviours	Unhealthy signs/behaviour
	General appearance		
	Movement		
	Appetite and feeding		
	Urine colour		
	Faeces and defaecation		
	Breathing		

(b) Common vices in farm animals

Type of livestock	Common vices	Management measures

Causes of poor health in livestock

Poor health in livestock is associated with several causes, including poor feeding, poor hygiene, poor housing, parasite infestation, injuries due to poor handling, and microorganisms or pathogens.

- Poor feeding:** Feeds and feeding play a significant role in providing good health. Inadequate feeding and poor-quality feeds can lead to nutritional disorders like cannibalism and egg-eating in chickens. Excessive feeding may lead to bloat, obesity in layer chickens, diarrhoea, or constipation.

- (b) **Poor hygiene:** Poor sanitation encourages the growth of microorganisms, which may infect the animals.
- (c) **Parasite infestation:** Internal or external parasites cause different effects on their host. Some of the effects caused by internal parasites, mainly gastrointestinal worms, compete with the host for food, reducing the host's appetite and causing ulceration and dehydration due to diarrhoea. These effects lead to poor health of the affected animal.
- (d) **Microorganisms/pathogens:** Pathogens of importance are bacteria, viruses, protozoa, fungi, mycoplasma, and rickettsia. Some modes of transmission or spread to infect healthy animals are contaminated water, feeds, pasture, air droplets, and bedding materials. Others include vectors and contact with the infected animal. When the pathogens come into contact with or enter the healthy animal's body, they cause different effects, including anaemia, haemorrhages, skin conditions, irritation, emaciation, loss of weight, and reduced production.
- (e) **Injuries due to poor handling:** Injuries are physical or traumatic conditions to the body tissues or organs caused by different sources. Slippery floors, cruel treatment by attendants, animal fighting, poorly constructed crushes, and accidents are sources of injury that may cause fractures, wounds, and other ill conditions to the animal.

Livestock disease management

In livestock disease management, two approaches are applied: curative and preventive. Curative methods include using drugs and chemicals to treat the disease. In contrast, preventive methods focus on measures that prevent the animal from coming into contact with the disease-causing agents or their vectors.

- (a) **Preventive measures:** Preventive measures used in managing livestock health include vaccination/immunisation, control of vectors, use of prophylactics, quarantine/isolation, sanitation, and proper nutrition.
- (b) **Curative measures:** These include:

- (i) Treating sick animals by applying drugs and chemicals to cure infectious diseases such as mastitis and pneumonia. Treatment helps to cure the animal and prevent the spread of such diseases.
- (ii) Deworming by drenching or injection (for controlling internal parasites) and dipping (for controlling external parasites) controls the parasites that may cause diseases and discomfort to animals.

Exercise 6.1

Use any learning resources, including library/internet search, video clips, experienced livestock keepers and experts to:

1. List any five plant protein feedstuffs used as ingredients in a mixture ration.
2. Outline any three common established pastures for livestock.
3. Describe how the following factors can cause illness to farm animals.
 - (a) Poor feeding;
 - (b) Poor hygiene;
 - (c) Microorganism; and
 - (d) Poor handling of animals.
4. Suggest how animal illness caused by the above factors in 1 (a-d) can be managed.
5. Discuss how animal housing can influence livestock production.

Breeding and reproduction management in livestock

Breeding and reproduction are essential phenomena for the existence and continuity of all living things. The breeding process to produce high-performing animals is a complex science performed by animal breeders. In this chapter, the roles of the livestock farmer are:

- (a) Selecting animals based on performance records of their dam, sire, and offspring;
- (b) Mating animals of the desired characters while ensuring they are not related;
- (c) Raising off-springs using the best management practices; and
- (d) Making replacement of animals from reliable, healthy stock of high performance.

Growth and maturity in farm animals

A newly born animal grows and develops various body organs and systems to an age when it is said to be sexually mature. At this age, it is said to have reached puberty, and under normal health conditions, it starts producing ova (for females) or sperms (for males). The reproductive cycle is a natural mechanism by which a mature animal undergoes the reproductive and non-reproductive phases. In mammals, the reproductive phase is also termed the oestrus cycle. It is divided into four stages: proestrus, oestrus, metestrus, and dioestrus, Tables 6.1 and 6.2 provide some details on the reproductive cycles, heat duration, ovulation, and gestation periods for various classes of livestock.

Table 6.1: Age at puberty for some farm animals

Animal species	Age at puberty (months)	The gestation period (average days)	Name of offspring	Act of giving birth
Cattle	10-18	281	Calf	Calving
Sheep	5-7	150	Lamb	Lambing
Goat	5-7	151	Kid	Kidding
Rabbit	4-6	30	Kitten	Kindling
Chicken	5-6 (varies with breed)	Brooding time 21 days	Chick	Hatching

Table 6.2: Lengths and duration of oestrus and time of ovulation in livestock

Animal species	Length of cycle (average days)	Duration of oestrus (average hours)	Time of ovulation (hours)
Cattle (cows)	20	16-24	10-14 after the onset of oestrus
Sheep (ewes)	16	30	1 before the end of the heat
Goat (does)	20	36-48	Near the end of the heat

Activity: 6.5

Visit the school farm or nearby livestock farms, make observations and inquiry on:

- (a) Signs/behavioural changes of a farm animal on heat; and
- (b) Management practices for the animal.

Livestock breeding systems

In livestock breeding, the animals with desired characteristics are selected and allowed to mate. In this way, the desirable genes from these animals are inherited, that is, transmitted to the offspring, making it perform similarly or even better than the parents. Proper livestock breeding ensures the continuity improvement of farm animal generation after generation. Thus, the main objectives of livestock breeding are to:

- (a) Obtain high-quality animal products, e.g., milk, meat, wool, and eggs.
- (b) Increase the production capacity of the animal in terms of quantity, such as milk yield, growth rate, weight gain, and number of eggs.
- (c) Increase the tolerance of an animal against diseases.
- (d) Increase animal adaptability to the environment.

Factors to consider in selecting an animal for breeding

For a breeding system to give good results, it has to go together with proper selection; hence, selection and breeding systems are also termed as tools for improving livestock's genetic makeup. Using these tools, different animal traits are identified, selected, and measured, and the best ones are used as parent animals. In the selection process, specific individuals are chosen to be the next generation's parents based on their desirable characteristics. The principle is to 'keep the best and cull the poorest.' On farms, various factors are considered when identifying the individuals to be selected, mainly based on observable and measurable traits. The factors include:

- (a) Age of the animal: Young animals are given priority because they have longer productive lives than old ones, and breeding efficiency declines with age.
- (b) Performance level: Animals with the highest production records should be selected first.

- (c) Health status: Healthy, disease-resistant animals, and free from abnormalities should be selected as they pass on to their offspring.
- (d) Quality of products: Selected animals should come from those that produce high-quality products.
- (e) Body conformation: Animals selected should have the desired conformation, e.g., wedge shape for dairy cattle and block shape for beef cattle.
- (f) Temperament: Select an animal with good behaviour; e.g., docility is desired for draught animals. Aggressive animals, especially bulls, should not be selected.
- (g) Prolificacy: Select animals that can produce large litter sizes (i.e., produce many offspring per parturition).
- (h) Adaptability: Select animals that can fit or cope with the prevailing environment.
- (i) Fertility: Select fertile animals that can breed regularly.
- (j) Mothering ability: Selected animals should be able to rear their young ones successfully to weaning age with good weight.

Techniques of breeding farm animals

Breeding farm animals involves identifying and selecting the animal to be bred to produce the next generation, whether male or female. Once a good quality stock has been identified, the next step will be determining the mating method used and the breeding season. The mating method can be done naturally or artificially, as described herein.

(a) Natural mating

In this method, a breeding male is used directly to mate the breeding female. Natural mating is done by allowing the male to detect and mate with the female in the field, in most cases while grazing, or by detecting the heat and bringing the male to mate. The former is pasture mating, while the latter is controlled mating. Figure 6.7 shows natural mating in cattle.



Figure 6.4: Natural mating in cattle

(b) Artificial insemination (AI)

Artificial Insemination (AI) refers to the process of depositing semen into the female reproductive organ by artificial means. AI is more common in cattle than in other livestock species. In cattle, the cow is inseminated only when it is observed to be on heat 10 - 14 hours after the onset of the heat period. Semen is collected from male animals by artificial means using appropriate facilities and materials. The portion of semen in diluted form or fresh is deposited into the uterus or uterine horn of the cow by mechanical means. Placing of the semen into the female is done timely to meet the ovulated egg for conception. The process is done under high hygiene to avoid contamination of the semen and the reproductive system. For successful insemination, the procedure should be performed by trained personnel. Figure 6.5 demonstrates AI in cattle.

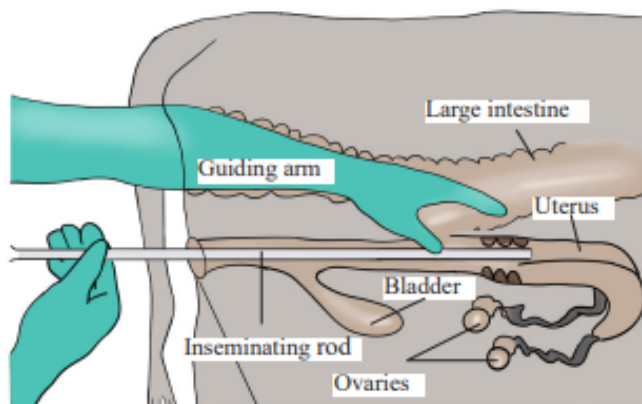


Figure 6.5: Demonstration of artificial insemination in cow

Activity 6.6

Visit any nearby AI centre/ study a relevant video clip on AI technology and:

- (a) Study on how AI is conducted; then,
- (b) Summarise your observation in your portfolio.

Exercise 6.2

1. Use any learning resource such as the library or internet search to explain the following terms as used in animal breeding and reproduction:
(a) Breeding (b) Reproduction (c) Oestrus (d) Oestrus cycle.
2. Outline any five signs of an animal on heat.
3. Describe four phases of oestrus cycle in a cow.
4. With reasons, suggest the best mating option for a farmer with only one bull and 20 cows on heat at the same time.
5. Identify any five materials/equipment used during A.I.

Handling and processing of livestock products and by-products

Usually, when an individual engages in livestock production, the ultimate aim is to sell the animals and/or their products and by-products. Animal products are the primary produce of the enterprise. They include live animals, meat, milk, and eggs. On the other hand, livestock by-products are the secondary produce of the enterprise. Animal by-products include hides/skin, horns, hooves, blood, offals, and manure. All livestock dealers, including those collecting animals from remote areas bringing them to the markets and processors of animal products, must abide by standards that govern animal transportation and handling practices to ensure their quality.

Handling of live animals

Before transporting animals for slaughter or rearing, the following actions should be undertaken:

- (a) Securing movement permits for the animals: When moving animals from one location to another, one must obtain a special certificate/permit from the responsible authorities. This is a compulsory regulation for all types of animals. This allows responsible authorities to inspect the animals'

health status and minimise the risks of spreading animal diseases. It also facilitates the control of the theft of animals.

- (b) Ensuring safety for both animals and their handlers: Animals sent to auctions or slaughterhouses must not be mistreated, as they may react in defence against discomfort and could harm themselves or their handlers. When animals are to be transferred at a distance for slaughter, they should be carried in special containers, cages, and vehicles (refer to Figures 6.6 and 6.7).
- (c) Ensuring comfort to the animal during transportation: Animals in transit should be provided with adequate space for standing or resting during transit and provided with food and water. Cattle, sheep, and goats that are herded on foot should be trekked within allocated stock routes where there are places for resting and clinical examination. For small animals like rabbits and chickens transported in cages, the cages must be of a size that shall not cause harm by squeezing or over-congestion. Chicken, for example, is very sensitive to overcrowding and heat stress.



Figure 6.6: Crates for transporting poultry

Source: <https://www.chickensandmore.com/wp-content/uploads/2021/01/Chicken-Crates.jpg>



Figure 6.7: Transportation of cattle

Source: <https://kb.rspca.org.au/wp-content/uploads/2020/05/livestock-transport-hot-weather-crop.jpg>

Activity 6.7

1. Through the library and internet search, find out the meaning, uses, and characteristics of the following, in livestock production:
 - (a) Slaughter slab;
 - (b) Slaughterhouse; and
 - (c) Abattoir.
2. Visit one of the following nearby structures in your community: slaughter slab/house/ abattoir and study on:
 - (a) Characteristics features of the structures
 - (b) Slaughtering procedures for the following animals:
 - (i) Ruminant animals;
 - (ii) Chicken; and
 - (iii) Rabbit.

Exercise 6.3

Use various learning resources, including library, internet searches, and resource persons, to answer the following questions:

1. Giving four examples in each case, differentiate livestock products from livestock by-products.
2. With examples, outline five characteristics of good quality packaging materials used for animal products.
3. Outline any four importance of processing livestock products.
4. Explain four various ways of enhancing the shelf-life of livestock products.
5. Identify five components of a slaughter (slab) house.
6. Describe the basic characteristics of a good vehicle for transporting livestock animals.

Livestock farm record keeping

Record keeping is necessary for the effectiveness and efficiency performance of any livestock production enterprise. A livestock register is a tool used to record and keep relevant information on animal farms. Some information includes the number of animals on the farm, animal identification number, date of birth, sire number, dam number, calf and its sex, date of calving, date of purchase, and date of sale/auction/death. Financial records (including costs and revenue) and health records are among the important information to be included in a farm register.

Activity 6.8

Visit a school farm or a nearby livestock farm and perform the following task:

- (a) Observe types of documents used in farm record-keeping;
- (b) Identify types of livestock records kept at the farm;
- (c) Study how each type of record is used in the management of the farm; then,
- (d) Prepare a report in your portfolio.

Exercise 6.4

1. What do you understand by the following terms?
 - (a) Sire; and
 - (b) Dam.
2. Why keeping farm records is important? Give five reasons.
3. Choose any livestock of your interest and identify key information to be recorded in the livestock register.



In your portfolio, summarise what you have learnt from this chapter.

Chapter Seven

Pig production

Introduction

In this chapter, you will learn about principles and practices underlying pig production, including housing, feeding, breeding, record keeping, health management, and marketing of pigs. The competencies developed will enable you to carry out general management practices in pig production enterprises.



Think

“A polyphagous livestock, with high capability to convert feeds into meat and giving birth twice a year with a high number of young ones.”

Concept of pig production

The wild boar (*Sus scrofa*) is the primary ancestor of the modern domestic pig. Wild boars are native to vast regions, including Europe, Asia, and North Africa. These highly adaptable animals allowed them to thrive in diverse environments. The domestication process involved humans selectively breeding wild boars for traits such as prolificacy, docility, meat production, and adaptability to human-provided environments.

Pigs also known as swine, have been vital to agriculture and human societies for millennia, providing meat, fat, and other products. Globally, pork is a primary source of animal protein, offering essential nutrients like high-quality protein, vitamin B, as well as iron, and zinc minerals. Figure 7.1 shows a lactating sow.



Figure 7.1: Lactating sow

As compared to other livestock, pig farming has the following relative advantages to livestock keepers:

- (a) **High reproduction rate (prolificacy):** Pigs can produce many piglets at once (an average of 10-12 piglets), and sows (mother pigs) can give birth 2 times a year. Farmers can quickly grow their herd and have more pigs to sell, leading to more income.
- (b) **Efficient use of feed:** Pigs are very good at turning feeds into body weight. Farmers can spend less on feed while getting pigs that grow fast and reach market weight quickly.
- (c) **Flexible on a diet:** Pigs can eat a variety of foods, including leftovers from the kitchen, farm crops, and other by-products. Pig keepers can use food that might otherwise be wasted, saving money on pig feed.
- (d) **High demand for pork:** Pork is a popular meat in Tanzania, farmers can quickly sell pigs and earn money, as there's always a market for pork.
- (e) **Quick growth:** Pigs grow fast and can be ready to sell in about 5-6 months under good management. Pig keepers do not have to wait for so long to make money from their pigs.

- (f) **Extra income:** Farmers can raise pigs for more money besides growing crops. Raising pigs helps farmers to have a steady income, even if their crops do not do well due to bad weather.
- (g) **Manure production:** Pigs produce manure, a natural plant fertiliser. Farmers can use pig manure to produce biogas, improve their soil, grow better crops, and spend less on industrial fertilisers, which are usually expensive. Manure from pigs can also be used for rearing insect larvae, which are used as protein, energy and fat sources for chicken and fish. An example of the insect larvae used in animal feed is the black soldier fly larvae.
- (h) **Job creation:** Pig farming provides jobs for many people, including young graduates. because pigs can be raised in peri-urban areas.
- (i) **Getting value-added products:** Pork can be turned into other products like sausages, bacon, and ham. These products can be sold at high prices, increasing the farmer's profit.

Activity 7.1

1. Use the library/ internet searches, slaughter houses, butchers or inquire from livestock keepers and experts on:
 - (a) Marketing of pig, pig products and by-products.
 - (b) Factors influencing pig production in Tanzania.
2. Summarise your observation in your portfolio.

Types of pig breeds

The selection of appropriate pig breeds is crucial in achieving specific production goals, such as meat quality, growth rate, and adaptability to environmental conditions. There are more than 600 pig breeds available globally. Common pig breeds in Tanzania include Yorkshire (Large white), Duroc, Saddleback, Landrace and indigenous pigs. There are also several hybrid breeds. Table 7.1 presents some common breeds kept in Tanzania.

Table 7.1: Some common pig breeds raised in Tanzania

Breed	Image	Description
Yorkshire/ Large white breed		<i>Origin:</i> England <i>Physical Features:</i> White with erect ears <i>Notable traits:</i> Known for excellent mothering ability, large litters, and high milk production. They have good growth rates and are often used in crossbreeding programs for their genetic contribution to improved production.
Duroc breed		<i>Origin:</i> United States <i>Physical features:</i> Red or reddish-brown with drooping ears. <i>Notable traits:</i> Known for fast growth rates, high feed efficiency, and quality meat with good marbling. Durocs are hardy and adapt well to various environments.
Landrace breed		<i>Origin:</i> Denmark <i>Physical features:</i> White with drooping ears <i>Notable traits:</i> include good mothering ability, large litter, and high milk production. They have long bodies, making them ideal for bacon production, and are often used in crossbreeding.
Saddleback breed		<i>Origin:</i> England <i>Physical features:</i> Black with a white belt, like Hampshire, but with drooping ears <i>Notable traits:</i> Have good mothering ability, adaptability to outdoor systems, and producing flavourful meat. They are hardy and have good foraging ability.

Activity 7.2

1. Use the library and internet searches to study on characteristic features of two more pig breeds
2. Visit a school farm or some nearby pig farm and perform the following tasks:
 - (a) Observe, and identify different breeds of pigs found on the farm;
 - (b) Comment on the observable features of the breeds on the visited livestock farm; and
 - (c) With reasons, suggest a breed of your preference for keeping.
3. Summarise what you have learnt from this activity in your portfolio.

Classes of pigs

Pigs are classified based on live weights, slaughter weights, and growth reproductive stage. Classes of pigs based on live weight are weaners, growers, and finishers. On the other hand, the classes of pigs based on slaughter weight are porkers, cutters, baconer, and heavy hogs. Based on growth and reproductive stage, they are classified as piglets, weaners, growers/finishers, breeders (breeding parents), pregnant sows/gilts, and lactating pigs.

Activity 7.3

1. Use various learning materials, including reference books, internet searches, and resource persons, to identify the characteristic features of pigs based on live-weight slaughter weights and reproductive stage.
2. In your portfolio, summarise what you have learnt from this activity.

Types of pig farming systems

Farming systems in pig production, depend on the number of pigs kept, and the production and management level, resources available, and environmental conditions. Generally, the common pig production systems are intensive pig farming, semi-intensive farming, and extensive farming. These are described hereafter:

(a) Extensive pig farming

In this farming system, pigs are raised in large outdoor areas with minimal confinement. They have access to open pastures (Figure 7.2), and their diet includes

natural forage like grasses, roots, and insects. Usually, they are supplemented with little feed. This system is often low-cost and requires minimal infrastructure. However, it is prone to diseases, parasites and predators and is not recommended as it also results in low productivity.



Figure 7.2: pigs kept under an extensive farming system.

Source: <https://www.thepigsite.com/articles/characterising-outdoor-pig-production-in-europe>

(b) Semi-intensive pig farming

This farming system combines indoor and outdoor management (Figure 7.3). Pigs are provided with shelter but also have access to outdoor areas. Farmers have more control over feeding and health management than extensive systems.



Figure 7.3: The semi-intensive pig farming system; sows feeding in troughs in a fenced unit.

Source: <https://www.fwi.co.uk/livestock/pigs/11-ways-pig-farmers-can-aim-for-net-zero>

(c) Intensive pig farming

In an intensive farming system, pigs are kept in confined, controlled environments such as barns or houses (Figure 7.4). All aspects of the pigs' environment, including feeding, temperature, and health, are closely managed. This system is highly industrialized and focuses on maximizing production efficiency.



Figure 7.4: An intensive pig farming system. The picture shows the gilts in the breeding unit.

Source: <https://www.thepigsite.com/articles/the-importance-of-a-good-gut-feeling-in-pig-production>

In some rural and peri-urban areas, a typical small-scale backyard pig farming is practised. With backyard farming, pigs are kept in small enclosures (Figure 7.5) or tethered in the yard. Farmers often use household food scraps and locally available feed.



Figure 7.5: Backyard farming system.

Source: <https://www.facebook.com/photo/?fbid=a.1640627276215475>

Activity 7.4

1. Make a simple survey on pig farming systems practiced in your community; to:
 - (a) observe the main features of each farming system practiced;
 - (b) identify the advantages and disadvantages of each identified system; and
 - (c) summarise your observations in your portfolio.
2. Adopt a farming system that suits better for you to raise some pigs at home/school.

Exercise 7.1

1. Using various learning resources such as the library, internet searches, and experience from your community discuss the level and status of pig production in your area.
2. Outline any four advantages of intensive system to extensive system of pig production.
3. Given an opportunity to engage in pig production, which farming systems would you opt for? Give reasons?
4. Explain the contribution of back yard pig production to the household income earnings.

Principles and practices of pig production

To raise any livestock profitably and sustainably, it is recommended that the principles and practices of livestock production be adhered to. The principles and practices of livestock production include proper housing, proper feeding, disease and parasite control, proper selection and breeding, proper processing and marketing of livestock products and by-products, and keeping pig farming records.

Housing of pigs

The pig house should have enough space for each pig to move around (Table 7.2 presents the optimum floor space requirement per age group of pigs). It should have good ventilation to allow fresh air and be kept clean to prevent diseases. It should also protect pigs from bad weather like rain, extreme heat, and cold. Several pig housing systems are designed to meet the needs of pigs at different

stages (piglets, growers, fatteners/finishers, adults, breeding boars, and lactating sows).

A proper house for pigs should have the following features:

- (a) Ability to protect pigs: A house for pig should be well constructed using durable materials to protect the animals from adverse weather conditions (rain, sun, wind) and protect them from theft and predators.
- (b) Adequate space: To allow pigs to move freely and access feed and water without crowding. It should have a separate area (pens) for different stages of growth.
- (c) Well-ventilated: To ensure a continuous supply of fresh air and remove excess heat, moisture, and harmful gases (e.g., ammonia).
- (d) Facilitate easy cleanliness and hygiene: The floors, walls, and equipment should be easy to clean and disinfect to maintain high hygiene standards.
- (e) Adequate lighting: Natural light through windows or skylights to support pigs' circadian rhythms. Alternatively, artificial lighting can be supplemented to ensure the proper light levels, especially in darker environments.
- (f) Feed systems should be designed to minimise feed wastage and ensure all pigs have access to feed.
- (g) Bedding and comfort: Use bedding materials (e.g., straw, sawdust) to provide comfort, absorb moisture, and reduce stress. Proper insulation and bedding help pigs regulate body temperature and stay warm.
- (h) Biosecurity measures: Access to the piggery should be restricted to prevent disease introduction. Footbaths, hand sanitizers, and disinfection points should be available to maintain cleanliness.
- (i) Emergency preparedness: Fire alarms and safety equipment should be installed to manage fire risks. Clear emergency plans, including those for health crises, natural disasters, and power outages, should be installed on the walls.
- (j) Efficiency and sustainability: Use resources (feed, water, energy) efficiently to reduce costs and environmental impact. Implement waste recycling or composting systems to utilise manure and minimise ecological impact.

Table 7.2: Optimum floor space requirement per age group for pigs

Class of pigs	Total floor space (m ² per pig)
Piglets/Weaners	0.5
Growers	1.0
Fatteners	2.0
Adults	2.5-5
Breeding boar	10
Lactating sow	7-10

Source: Sibuga *et al* (2004)

Activity 7.5

Pay a visit to school and nearby pig demonstration farms (such as livestock research centers, livestock institutes, and outstanding pig keepers) to perform the following tasks:

- Observe the general set-up of the piggery unit and types of houses for various classes and farming systems of pigs; then,
- Summarise your observations in your portfolio.

Piggery

Like any other animal-related enterprise, pigs require proper housing, feeding, health care, breeding, record keeping, and other management practice. All these practices must be carried out at an organized pig production place called piggery. A piggery is a farm or a specialised area where pigs are raised. A piggery comprises of:

- Feeding area: The feeding place includes the space for feeds and water.
- Resting place: Pigs like to rest most of the time and, therefore, need a good resting space.
- Space for dunging and exercising: Pigs require an area for dunging and exercising; exercising makes them strong and fit, which is particularly important for breeding males.
- Drainage system: Pigs are clean animals if provided with enough space. A draining system is required to collect and take away urine and dung to the disposal point.

- (v) Feed store: A piggery should contain a store for feedstuffs.
- (vi) Space/office for records: There should be a place for recording farm information such as feeds and feeding, diseases and parasites, breeds and breeding, and marketing.
- (vii) Space for tools and equipment: The piggery unit should have a place for storing farm tools and equipment.

Exercise 7.2

1. Outline the harmful effects of poor housing on different classes of pigs.
2. Explain the uses of any four equipment found in piggery unit.
3. Discuss how drainage system affect pig production.

Management of different classes of pigs

In the piggery, pigs should be separated into different sections according to their age or reproductive stage.

Management of piglets

Piglets are young pigs from birth to weaning. A group of piglets born at the same time from the same sow (mother) is called litter. Good management practices are crucial for the piglets to grow well and reach the disposal stage. The managerial practices include colostrum provision and cross-fostering of piglets, tail docking, iron injection, identification, castration, and teeth creeping. These practices are elaborated hereunder.

(a) Colostrum provision and cross-fostering of piglets: The young piglets must get colostrum within 2 to 3 hours after farrowing for high viability. Tying and dipping of the navel cord in the iodine solution should be done immediately after birth to prevent infection through the navel. Cross-fostering should be done shortly after the sow completes farrowing. Cross-fostering involves even the number of piglets from different sows. It is aimed to ensure all the piglets can access teats to suckle and ensure that the sows farrowing few piglets below the average have piglets to nurse.

(b) Tail docking: Tail docking is performed to prevent tail biting among pigs, which can lead to injuries and infections. The procedure involves removing a portion of the piglet's tail using a heated docking iron or surgical scissors. Figure 7.6 (a) and 7.6 (b) shows a docking process and docked piglets respectively.



Figure7.6 (a): Docking in piglets



Figure7.6 (b): Docked piglets

(c) Iron injection: New-born piglets are naturally low in iron, which can lead to anaemia. Iron injections are given to prevent iron deficiency and promote healthy, growth and development. An iron supplement is injected into the muscle, typically in the neck (Figure 7.7) or ham area, within the first 3 to 4 days after birth. Oral iron supplements can also be used as an alternative in some systems.



Figure 7.7: Iron injection to a piglet

(d) Identification: Identification in animals facilitates record keeping and symbolises ownership, among many advantages. It is a crucial practice in differentiating pigs. There are different types of identification methods used in pigs. The methods include ear tattooing, ear notching, and ear tagging.

- (e) **Teeth clipping:** Teeth clipping is performed on a new-born piglet shortly after birth. The procedure involves trimming or clipping the sharp needle-like teeth, also known as needle teeth (Figure 7.8). Needle teeth may cause injuries to the sow's teats, other piglets, or even the caretaker.



Figure 7.8 Clipping needle teeth using tooth clipper.

- (f) **Castration:** Male piglets are often castrated to prevent aggressive behaviour and the development of boar taint, an unpleasant odour, and flavour in the meat of mature male pigs. Castration can also mean controlling unwanted breeding. Surgical castration involves the removal of the testicles or crushing the spermatic cord typically performed within the first week after birth when piglets are less sensitive.
- (g) **Creep feeding:** Creep feeding is the practice of providing supplemental feed to nursing piglets during the transition period from milk to solid feeding. It ensures adequate nutrition and promotes growth and development. Creep feeding typically begins when piglets are around 1 to 2 weeks old and continues until weaning. The feed is usually high in energy, protein, essential minerals, and vitamins to meet the piglets' growth needs.
- (h) **Weaning:** Weaning is the management practice of separating piglets from their mother (sow) by gradually stopping piglets from suckling the sow's milk and starting feeding solids feeds. To avoid stress to piglets, the usual practice is to remove the sow from the pen and leave piglets in the farrowing

room. Depending on the piglets' health conditions, in commercial farms weaning is normally done at 3 to 4 weeks of age. In local production systems, weaning can be done later when piglets are around 8 weeks old. Shortly before weaning, gradually reduce milk access while increasing solid feed to help ease the transition. After weaning, piglets are moved to a nursery facility with appropriate temperature, ventilation, and space to reduce stress and promote adaptation.

Activity 7.6

1. Visit a school farm and some nearby pig farms and carry out the following task:
 - (a) Observe various piglets management practices performed in the farm;
 - (b) Under the guidance of a farm manager/experienced livestock keeper, participate in carrying out some of the management practices.
2. Prepare a report in your portfolio.

Exercise 7.3

1. Describe how the following practices are conducted in the pig farm:
 - (a) Castration;
 - (b) Teeth clipping in piglets;
 - (c) Identification in pigs; and
 - (d) Weaning.
2. Discuss how castration improves meat quality in farm animals.
3. Describe how you will handle the case of a sow that rejects piglets introduced from another sow which had died immediately after farrowing.

Management of weaners, growers, and fatteners

Effective management of weaners and fatteners (pigs raised for meat) ensures their health, growth, and overall productivity. The following practices are employed;

- (a) Weaners and growing pigs should be fed with small amounts multiple times daily for better digestion and nutrient absorption.

- (b) Adjust feeds to match growth rates.
- (c) Provide enough space to allow maintenance of hygiene.
- (d) Keep the feeding area clean and provide fresh feed and water daily to promote health.
- (e) Reduce stress by maintaining a calm environment.

Management of lactating sows

Good management of lactating sows should start a week before farrowing (giving birth). One week before farrowing, the sow should be dewormed to kill the worms, and broad-spectrum de-wormers should be used to control both external and internal parasites. Likewise, feeding should be reduced to approximately one-half of the sow ration. Two days before giving birth, feed the sow with laxative feeds such as brans and molasses to prevent constipation. Proper feeding of sows is essential, as poorly fed sows may eat their piglets after farrowing.

For the sow to produce sufficient milk for the piglets, it should be adequately fed, and its diet must be balanced with all nutrients. On the farrowing day, provide a sow with plenty of water and a small amount (about 0.25kg) of bran, molasses, or other laxative feeds. On the farrowing day, feed the sow 0.5kg of bran or molasses, and in the following days, increase the feeds by 1 kg of feed per day until the required amount of feed is attained.

Feed each sow with a minimum of 3kg feed per day to satisfy its body requirements, and for each piglet, add 1/3 kg of feed (Table 7.3). Alternatively, feed the sow ad libitum. At weaning time, gradually reduce feed amounts by 1kg until only 2 kg per day.

Table 7.3: Restricted feeding plan of a sow

Number of piglets in the litter	Amount of feed per day (kg)
3	4
6	5
9	6
12	7
15	8

Activity 7.7

1. Visit your school farm/family farm or nearby pig farms and study on factors affecting the performance of growing pigs;
2. Suggest management measures to the factors affecting the performance of the farms you have visited.
3. Record your observation in your portfolio.

Management of pregnant sows

For the sow to produce many viable and strong piglets, caring for the sow should start before mating (during flushing). Flushing put the sow's body in good condition, allowing the sow to farrow large litters of heavy piglets. Flushing should be practised one week before Mating (meeting a boar).

During pregnancy, feed a sow with 2 -2.5 kg of feed per day and increase the amount to 3 kg during the last three weeks of pregnancy.

Parturition

The action of giving birth in pigs is called farrowing. Care must be taken during the farrowing to avoid the piglet's death if anything goes wrong. The following preparations must be done before and during farrowing:

Preparation of the farrowing pen

The farrowing pen should have the following facilities:

- (a) Guard rails or farrowing crates: The guard rails protect piglets from being crushed by their mother when it is laying. If a farrowing crate is provided instead, it confines the sow from moving and turning around and crushing the piglets.
- (b) An artificial source of heat (artificial heating device) for the piglets should be placed/installed in the pen if heat does drop below 20 °C.
- (c) Creep feed: Creep feed must be provided in the farrowing crate in the pen for the piglets to access.
- (d) Bedding materials: Provide bedding materials on the pen floor, such as straws, husks, hay sawdust, or dry banana leaves.

Remove the sow from the herd four days before farrowing, wash it thoroughly with disinfectants, and move it to the farrowing pen.

Signs of farrowing

When a sow is approaching farrowing, will show some of the following signs:

- (a) The sow becomes restless;
- (b) The sow becomes nervous and uneasy;
- (c) The Sow urinates frequently;
- (d) The vulva becomes red and swollen;
- (e) Sometimes, whitish mucous discharges may come out of the vulva;
- (f) The abdomen and mammary tissue protrude; and,
- (g) The sow starts secreting milk from the teats.

Care of the sow and piglets during farrowing

Careful observation of the sow at farrowing is necessary to observe any abnormalities that may lead to the death of piglets. If the sow is restless, remove each piglet after it is born. Once the process is completed, return the piglets to the sow. Usually, the piglets will start to suckle immediately after farrowing is over.

Activity 7.8

1. Inquire from experienced pig livestock keepers, library and internet searches on the possible consequences that may happen if farrowing takes place in absence of an attendant.
2. Summarise what you have learnt from this activity in your portfolio.

Feeding and feed formulation for pigs

Like other farm animals, pigs need the right feed to grow strong and healthy. Pigs should be fed a balanced diet with energy-rich foods like grains, proteins like soyabean meal, and vitamins and minerals. Fresh, clean water should always be available. Feeding should be done regularly, and the amount of food should be adjusted as the pigs grow. Formulating a pig ration requires knowledge of the nutritional needs of pigs at various stages of growth and the availability of feed resources.

Activity 7.9

1. Pay a visit to your school farm/nearby pig farm or livestock expert and learn about:
 - (a) Feed formulations for different classes of pigs; and
 - (b) Participate in formulating various feed rations for various classes of pigs;
2. Write a summary of what you learned from this activity in your portfolio.
3. Adopt and apply appropriate feed formulation skills in the school/family farm.

Exercise 7.4

1. Define the following terms as used in pig feeding:
 - (a) Feed;
 - (b) Feed formulation; and
 - (c) Balanced ration.
2. Why it is important to feed a pregnant sow a balanced ration?
3. In a tabular form:
 - (a) Identify different types of feed materials used in feed ration formulation;
 - (b) For each identified feed material in 1 (a):
 - (i) outline the primary nutrient contained in the feed.
 - (ii) suggest one alternative feed source.

Breeding practices

Good breeding practices help produce healthy piglets and ensure the next generation is strong. A healthy pig with good traits (like fast growth or strong immunity) should be chosen for breeding. The pregnant sows (mother pigs) should be cared for, by providing proper food and a comfortable place to give birth. Effective management of the breeding herd, which includes breeding boars and gilts, is crucial for ensuring the productivity and sustainability of pig production on the farm.

Characteristics of breeding pigs (boars, sows, and gilts)

Breeding pigs (boars, sows, and gilts) should possess the following features:

- (a) **Genetic quality:** Select breeding animals from lines with desirable traits such as high fertility, fast growth rates, feed efficiency, disease resistance, and good carcass quality. Ensure that boars and gilts come from well-documented lineages aligning with the farm's production goals.
- (b) **Physical conformation:** Both boars and gilts should have strong, well-developed bodies with balanced structures. Boars should have sturdy legs and feet to support their weight during mating, while gilts should have well-aligned legs and feet to support them during breeding and nursing. Gilts need a well-developed udder with at least 12 functional teats to ensure adequate milk production for future litters.
- (c) **Reproductive performance and maturity:** Boars should have fully developed and healthy reproductive organs, high libido, and good semen quality. Gilts should reach sexual maturity around 6-8 months, with regular estrus cycles, and be bred when they get around 135 kg to ensure they are mature enough to carry a litter successfully.
- (d) **Temperament:** Both boars and gilts should have a calm but active temperament. Boars that are too aggressive can cause injuries, while overly passive boars may not mate effectively. Gilts with a calm temperament are better mothers.
- (e) **Health and hygiene:** Regular health checks, vaccinations, proper nutrition, and hygiene are essential for boars and gilts. Both should be free from diseases and genetic defects.
- (f) **Longevity:** Select breeding animals with the potential for long reproductive lives. Boars should produce many litters over several years, and gilts should be capable of producing multiple large litters with low piglet mortality rates.

Management of boars and mating

Before mating, the sow and gilts should be flushed to prepare them for mating. Flushing involves increasing the feed intake of sows or gilts, particularly those in poor body condition, before breeding. The goal is to enhance their overall health and reproductive efficiency, leading to higher conception rates and larger litter sizes. Flushing typically starts 2-3 weeks before mating and continues throughout the breeding period. This ensures that sows and gilts are in optimal body conditions when bred.

Reproduction performance in pig production is influenced by many factors, including the boar-to-sow/gilt ratio, mating duration, and the breeding system employed. Natural mating (Figure 7.9 (a)) requires careful management of boars to avoid overuse, while Artificial Insemination (AI) (Figure 7.9 (b)) offers several advantages in terms of genetic improvement, disease control, and efficiency. However, the system requires skilled personnel, semen availability, and AI facilities.



Figure 7.9 (a): Natural mating

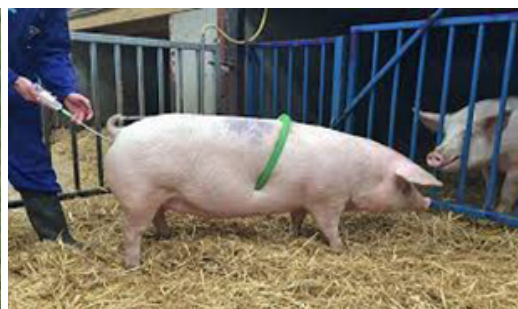


Figure 7.9: (b)) Artificial insemination

Activity 7.10

1. Visit any nearby demonstration farms, agricultural institutes/research centres, AI centres and or use video clips, internet search/ library to learn on:
 - (a) artificial Insemination;
 - (b) natural mating practices;
 - (c) flushing; and
 - (d) steaming-up.
2. Prepare a report in your portfolio.

Exercise 7.5

1. Give the mating ratio for commercial pigs.
2. In a tabular form, differentiate between natural mating and artificial insemination.
3. Briefly explain the meaning and importance of the following practices in pig production:
 - (a) Flushing; and
 - (b) steaming-up.
4. What are the effects of a pregnant sow farrowing in a non-farrowing pen?

Prevention of diseases and parasites

Healthy pigs grow better and produce optimally. Preventing and treating diseases is critical to successful pig production. Pigs should be checked regularly for signs of sickness. They should be vaccinated against common diseases and dewormed to eliminate parasites. Keep their living area clean and isolate sick pigs to prevent disease spreading. Effective control of common diseases and parasites in pigs is essential for maintaining herd health, optimizing productivity, and ensuring animal welfare.

Activity 7.11

Visit a nearby farm or do an internet search about the common parasites and diseases affecting pigs in Tanzania. Then, in a tabular form, indicate the symptoms of the disease, causative agent, and the control measures.

Handling, processing, and marketing pig products and by-products

Handling, processing, and marketing pig products and by-products are essential for the pig farming industry. The following are the steps to ensure that pig products are safe, high-quality, and reach the market effectively.

- (i) **Slaughtering:** Pigs should be slaughtered in special structures such as slaughter houses and abattoirs where humane practices are followed to minimise stress and ensure the welfare of the animals. Proper slaughtering techniques are vital for producing high-quality pork.
- (ii) **Meat processing:** Once slaughtered, the pig carcass is divided into various cuts like loins, hams, and ribs. These cuts are further processed into products such as bacon, sausages, and ham. Processing involves curing, smoking, and packaging, which preserve the meat and enhance its flavour. These processes also extend the shelf life of the products.
- (iii) **By-product utilisation:** Besides meat, pigs provide several valuable by-products. The skin can be processed into leather, fats are rendered into lard or used in industrial applications. Bones and blood can be converted into animal feed or fertiliser. Effective utilisation of these by-products reduces waste and maximizes the economic return from pig farming.
- (iv) **Quality control:** To maintain product quality, strict hygiene and storage protocols should be followed throughout handling and processing. Processing facilities should be kept clean, and products should be labelled and stored correctly to prevent contamination and spoilage.

- (v) **Marketing:** Marketing is a crucial aspect of the pig production industry. It involves product packaging attractively, setting competitive prices, and promoting them to consumers. Farmers and processors may market their products directly to consumers through local markets or supply retailers, restaurants, or export markets.

However, in Tanzania, live pigs are the most commonly traded. Therefore, you need to identify the market demands for piglets, live slaughter pigs, carcasses, and processed products and develop your marketing channels. Work out the most appropriate based on costs and profitability. Use social media and other appropriate promotion channels.

Exercise 7.6

1. Explain how to select piglets for breeding from a litter.
2. Explain how to assess the quality of a sow before breeding.
3. Describe the characteristics you would look for when selecting a boar for breeding.
4. Describe how to perform a routine health check on a pig.
5. In pig farming, stresses are inevitable and may have different sources. Analyse some stresses that can be caused by:
 - (a) Poor feeding/nutrition;
 - (b) Poor housing;
 - (c) Poor sanitation; and
 - (d) Poor watering.

Record keeping in pig farming

Different types of records are kept on a pig farm. Important information includes financial, feed, breeding, and stock (herd) records. For the sows, the following details must be included: the number of days between weaning and oestrus, weaning day, weaning to service interval, piglets born/litter size and weaned per litter, weaning age/weight and litter number. Such information is necessary to help identify managerial problems in the pig farm, accurately detect the signs of heat, and schedule insemination or mating at the optimum time.

Activity 7.12

1. Visit a school farm or a nearby pig farm and inquire/study on:
 - (a) types of pig records kept at the farm; and
 - (b) how each type of record is used in the management of the farm.
2. Prepare a report in your portfolio.



Write a summary of what you have learnt from this chapter in your portfolio.

Chapter Eight

Rabbit production

Introduction

In this chapter, you will learn about rabbit production principles and practices, including rabbit breeds, rabbit housing, feeding, breeding, record keeping, health care, and marketing of rabbits. The competencies developed will enable you to perform general management practices in rabbit production enterprises.



Think

“A four-legged animal whose meat tastes like chicken.”

Importance of keeping rabbits

People may choose to keep rabbits for various purposes. Some of them decide to keep rabbits for meat production. Choose breeds known for their rapid growth, high feed conversion efficiency, and good meat quality, such as New Zealand and California white. If one is interested in fur, one may choose breeds with high-quality pelts, like the Rex or Angora. Figure 8.1 presents some rabbits in a group pen.



Figure 8.1: Rabbits in a cage

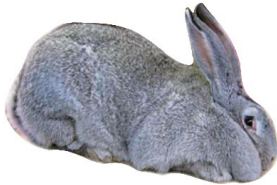
Raising rabbits can be beneficial for several reasons, especially for young farmers, families, and communities. The following are some relative advantages of keeping rabbits.

- (a) **Source of high-quality meat:** Rabbit meat is high in protein, low in fat, and contains essential vitamins and minerals. It's a healthy option for those looking to maintain a balanced diet. Rabbit meat is one of the leanest meats available, making it a good choice for those who want to eat healthy.
- (b) **Income generation:** There is a growing demand for rabbit meat in many areas. Raising rabbits allows one to sell the meat to local markets, restaurants, or directly to consumers, providing a steady source of income. Rabbits require less space, food, and care than larger livestock, making them an affordable option for small-scale farmers or those with limited resources.
- (c) **Highly prolific:** Rabbits have a short gestation period of about 30 days and can produce large litters, which helps to increase the stock and productivity quickly. Unlike other animals, rabbits can be bred yearly, providing a consistent meat supply and income.
- (d) **Easy to raise:** Rabbits are relatively easy to care for. They don't need much space, and their diet mainly consists of hay, fresh greens, and pellets, some of which are easy to provide. Rabbits can be raised in various environments, including small backyards, making them suitable for rural and urban settings.
- (e) **Educational value:** Raising rabbits can be an excellent way for young people to learn about responsibility, animal care, and essential farming practices. It's a hands-on experience that teaches necessary life skills. Caring for rabbits can also be a practical introduction to biology and science, as it involves understanding animal behaviour, nutrition, and reproduction.
- (f) **Environmental benefits:** Rabbits efficiently convert kitchen scraps and garden waste into high-quality protein, reducing food waste. Rabbit manure is an excellent natural fertiliser. It is rich in nutrients and can be used directly in gardens to improve soil fertility and boost crop production. Rabbit urine can also be used as a foliar fertiliser and insecticide for crop protection.
- (g) **Fur production:** In addition to meat, some rabbit breeds produce high-quality fur that can be sold or used to make clothing and other products, adding another revenue stream.

There are many breeds of rabbit kept worldwide. Rabbit breeds serve various purposes, from meat to fur production. Other breeds are used as pets and show

animals, making them versatile and valuable in different contexts. The description of some common rabbit breeds, each with its unique characteristics, is explained in Table 8.1

Table 8.1: Some common breeds of rabbits

Breed	Image	Description
New Zealand white		<i>Origin:</i> United States <i>Physical features:</i> Medium to large size, white fur, pink eyes, and a muscular build. <i>Notable traits:</i> Fast growth and high meat yield; often used in commercial meat production.
California white		<i>Origin:</i> United States <i>Physical features:</i> White fur with dark markings on the ears, nose, paws, and tail; medium to large size. <i>Notable traits:</i> Excellent meat breed; calm temperament and good mothering abilities.
Flemish giant		<i>Origin:</i> Belgium <i>Physical features:</i> It is large, weighing up to 6 kg or more, has a long body and large ears, and comes in various colours. <i>Notable traits:</i> It is friendly and docile, one of the giant rabbit breed used for meat and fur.
Chinchilla		<i>Origin:</i> France <i>Physical features:</i> It is medium-sized, with dense, soft fur and a distinctive grey colour, resembling a chinchilla. <i>Notable traits:</i> A bred for their fur; also used for meat and as pets.

Angora		<i>Origin:</i> Turkey <i>Physical features:</i> It is medium-sized, has long, soft wool covering the entire body, and requires regular grooming. <i>Notable traits:</i> It is primarily bred for its wool which is used in making yarn. Due to its dense fur, it requires special care.
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Principles and practices of rabbit production

The principles and practices of rabbit production include proper housing, proper feeding, disease and parasite control, proper selection and breeding, proper processing and marketing of livestock products and by-products, and keeping farm records.

Selection of rabbits for farming

When starting a rabbit farming venture, selecting the right rabbit parent stock (breeding stock) is vital for success. The following are the key factors to consider when buying rabbit stock:

- (a) **Breed selection:** Ensure the breed chosen is well-suited to the local climate. Some breeds may struggle in extreme temperatures (very hot/cold). Consider the breed's temperament, especially if one plans to handle the rabbits frequently or wants to raise them as pets.
- (b) **Health status:** Purchase rabbits from a reputable breeder who can guarantee that the rabbits are free from common diseases. Healthy rabbits should have clear eyes, clean ears, and a glossy coat. Ask for records of vaccinations and past medical treatments to ensure the rabbits have been appropriately cared for. Inspect the rabbits for signs of parasites, such as fleas or mites, which can affect their health.
- (c) **Age and maturity:** It's often recommended to purchase rabbits between 8 to 12 weeks old. At this age, they are weaned and can adjust to a new environment more quickly. If buying breeding stock, ensure the rabbits are

of the right age for breeding (usually around 5-6 months old for does and slightly older for bucks).

- (d) **Physical characteristics:** Look for rabbits with a strong, well-muscled body, straight legs, and good bone structure. These traits indicate good genetics and health. Depending on the purpose, choose rabbits within the desired weight range for their breed, ensuring they meet breed standards.
- (e) **Source of stock:** Buy from a trusted and reputable breeder with a good track record. They should be knowledgeable and willing to provide information on the care and history of the rabbits. If possible, visit the farm where the rabbits are raised. Purchasing from animal markets can be risky due to the potential exposure to diseases, as their health is not guaranteed.
- (f) **Genetic background:** If one plans to breed rabbits, knowing their pedigree is essential to avoid inbreeding and to ensure desirable traits are passed on. Consider the genetic characteristics of the rabbits, such as growth rate, litter size, and mothering ability, which are essential for a productive rabbitry.
- (g) **Cost:** Compare prices from different breeders to ensure getting good value for money. However, don't compromise on quality for a lower price. Consider the cost of transportation, potential veterinary checks, and any necessary housing adjustments when buying new stock.
- (h) **Long-term plans:** Consider the rabbitry's long-term sustainability. Choose rabbits that will provide a strong foundation for building a healthy and productive herd. If one wishes to expand, one should consider starting with a few high-quality rabbits that can breed to increase the stock over time.
- (i) **Support and guidance:** Purchase from a breeder who offers ongoing support and advice. This can be invaluable, especially if one is new to rabbit farming. Choose a breeder who can provide or recommend resources for learning more about rabbit care, breeding, and management.

Activity 8.1

1. Pay a visit to your school or nearby rabbit farm and perform the following tasks:
 - (a) Observe and identify the rabbit breeds available on the farm;
 - (b) Comment on the observable features of the breeds on the visited livestock farm;

- (c) Observe and practise some management practices done at the farm; and,
 - (d) Suggest measures that might be made on the farm to improve its productivity.
2. Summarise your observations in your portfolio.

Handling rabbits

Proper handling of rabbits is essential to ensure their well-being, prevent injuries, and make management tasks more manageable. Handling procedures for adults will differ from those for young rabbits because of their delicacy. The basic handling process in adults and young ones should consider a calm approach, correct lifting, and reduced stress.

Rabbit housing and the equipment

Proper housing and equipment are essential for successful rabbit farming, ensuring the rabbits' health, safety, and productivity. The following is a detailed explanation of the housing and equipment needed for keeping rabbits.

Types of rabbit housing

Hutches are individual or small group cages, usually made from wood or wire, elevated from the ground (Figure 8.2). They are ideal for small-scale rabbit farming or backyard setups. They provide shelter from predators and adverse weather conditions.



Figure 8.2: An example of a rabbit hutch

Source: (<https://www.factoryfast.com.au/products/large-rabbit-hutch-with-base-chicken-coop-2-storey-guinea-pig-pet-cage-house-1>)

Cages: Wire or metal cages are often used in commercial rabbit farming (Figure 8.3). Cages are suitable for larger operations where many rabbits are kept. Cages are easy to clean, have good ventilation, and can be stacked to save space.



Figure 8.3: An example of a rabbit cage

Source: (<https://www.amazon.com/AJAAS-Indoor-Rabbit-Cage-142x111x39cm/dp/B0B5L4Y7XS?th=1>)

Rabbitry (Enclosed building): This is a dedicated building that houses multiple cages or hutches. It is common in larger-scale rabbit farming operations. It protects from extreme weather and predators and allows better environmental control (temperature, humidity, light).

Characteristics of a rabbit house

- (i) Each rabbit should have enough space to move, stretch, and rest comfortably. Typically, a cage should be at least 3-4 times the size of the rabbit.
- (ii) Good ventilation is crucial to prevent respiratory problems. Ensure the housing allows for adequate air circulation without exposing rabbits to drafts.
- (iii) Rabbits thrive well in temperatures between 10°C to 21°C. Housing should protect them from extreme heat or cold.
- (iv) Provide natural or artificial lighting, mimicking a natural day-night cycle. Avoid direct sunlight, which can overheat the rabbits.
- (v) Ensure the housing is secure and elevated to prevent access by predators like dogs, cats, or rodents.
- (vi) Rabbit house can have wire mesh or solid flooring. Wire mesh flooring is common in cages, allowing droppings to fall through and keeping the living area clean. Ensure the mesh is not too wide to avoid injuring the rabbit's feet. Solid flooring is used in hutches, often covered with litter (straw or wood shavings.)

Equipment and facilities needed for keeping rabbits

The following are the common equipment required in the rabbit house.

- (i) **Feeders:** These are containers or trays designed to hold rabbit pellets or grains. They can be made of concrete, metal or plastic materials. Feeders can be attached to the cage. They should be easy to clean, durable, and designed to prevent food wastage.
- (ii) **Hay racks:** Wire racks or bins hold hay, a crucial part of a rabbit diet. They help to keep hay off the ground to avoid contamination.
- (iii) **Water bottles/bowls:** Bottles with sipper tubes that attach to the side of the cage. Alternatively, heavy, tip-resistant bowls made of concrete (refer Figure 8.1) may be used. Ensures rabbits have access to clean, fresh water at all times.
- (iv) **Nesting boxes:** Boxes placed in the hutch or cage where does (female rabbits) can give birth and care for their young. Often made of wood or metal, filled with straw or soft bedding. Provides warmth and security for new-born kits.
- (v) **Litter trays:** These trays or pans are placed under the cages to collect droppings. It makes it easier to clean and manage waste.
- (vi) **Brushes:** Regular brushing helps maintain a rabbit's coat and prevents matting, especially in long-haired breeds.

Carriers

Portable cages or boxes are used to transport rabbits safely. It is essential for moving rabbits to markets, vets, or new housing.

Activity 8.2

1. Pay a visit to various nearby rabbit farms to observe the following:
 - (a) different rabbit behaviours;
 - (b) rabbit handling/ processes (steps to follow); and
 - (c) housing systems used in rabbit production.
2. Record your observation in your portfolio.

Exercise 8.1

1. Outline the essential characteristics of rabbit cages/hutches.
2. Explain how will you handle a rabbit bought from a friend (5km away).
3. Outline uses of the following rabbit by-products in the community:
 - (a) urine;
 - (b) fur; and
 - (c) droppings.
4. Explain the:
 - (a) likely effects if, a rabbit kindles out of the kindling box. Give at least three points.
 - (b) good practices of handling kitten kindled outside the kindling box.

Feeds and feeding of rabbits

Feeding rabbits involves providing a balanced diet that meets their nutritional needs for growth, reproduction, and overall health. Rabbits feed on various feed types, including forages, grasses/legumes (fresh/hay), horticultural products, farm crops, grains, concentrates, and commercial rabbit feeds. Hay is the most crucial part of a rabbit's diet, making up about 70-80% of their daily food intake. It provides the necessary fibres for proper digestion, helps wear down their constantly growing teeth, and keeps their gut healthy. Supplements are generally not required if the rabbit is on a well-balanced diet, but they may be used in specific situations, such as during pregnancy, lactation, or illness. Supplements can be in the form of mineral licks or probiotics.

In addition, and in a unique way, rabbits tend to eat their soft faecal pellets (caecotropes). This behaviour is referred to as caecotrophy/night droppings. Caecotropes are rich in nutrients (protein and Vitamin B) and less in fibre, thus making them good nutrient supplements for rabbits.

Feeding schedule

Rabbits thrive on a consistent feeding schedule, typically twice daily (morning and evening).

A proper feeding regime ensures rabbits remain healthy, active, and productive.

Special feeding considerations

Young rabbits (kits): Kits should start nibbling on hay and pellets around three weeks of age, with complete weaning by 6-8 weeks. Initially, Alfalfa hay should be offered, and other feeds should be introduced as they mature.

Pregnant and lactating does: Pregnant and lactating does require a diet higher in protein, calcium, and energy. Increase the quantity of pellets and provide Alfalfa hay. However, overfeeding pellets can lead to obesity and digestive problems. Insufficient hay can cause dental issues and digestive blockages. Avoid feeding starchy vegetables (like potatoes) and sugary fruits in excess, as they can cause digestive upset.

Monitoring health: Regularly monitor rabbits eating habits, weight, and droppings. Any significant changes could indicate health issues and should be addressed promptly. Ensure the rabbit teeth are healthy by providing ample hay and occasionally offering safe chew toys or branches (like wood) to help wear down teeth.

Activity 8.3

1. Pay a visit to your school farm or a nearby rabbit farm, observe:
 - (a) feeding and drinking equipment;
 - (b) feeds and feeding routine of rabbits; and
 - (c) effects caused by poor feeding to rabbits.
2. Summarize the information in your portfolio.

Exercise 8.2

1. Identify different types of feeds used in rabbit production.
2. Referring to the feed types identified in question 1,
 - (a) explain essential characteristics for each feed type
 - (b) identify limitations and suggest solutions for each.
3. When is considered the right time for providing supplement feed to kittens?

Breeding in rabbit

Breeding and pregnancy in rabbits involve steps and considerations to ensure successful reproduction and healthy offspring. Select healthy rabbits that are free from disease and have good physical condition. Both males (bucks) and females (does) should have desirable traits, such as good size, conformation, and temperament. Rabbits are usually bred when they are 6-8 months old, but this can vary depending on the breed.

Mating process: Introduce the doe to the buck's cage or pen. It is recommended to place the doe in the buck cage to avoid territorial behaviour exhibited by a doe.

Mating signs: The doe will often display signs of receptivity, such as sitting still and allowing the buck to mount. If successful, mating usually takes only a few minutes. After mating, observe if the doe displays a "lordosis" position (arched back), which indicates successful mating. It's often advised to reintroduce the doe to the buck 12-24 hours later to ensure successful breeding.

Pregnancy confirmation: Pregnancy in rabbits can be confirmed about 10-14 days after mating by gently palpating the doe's abdomen. Alternatively, a veterinarian can perform an ultrasound for early confirmation. A pregnant doe may show increased appetite, a slightly enlarged abdomen, and behaviour changes such as nesting.

Pregnancy (gestation) in rabbits: Rabbit pregnancy lasts about 28-32 days, with an average of 31 days. Ensure the doe has a comfortable, clean, and safe environment. A balanced diet with adequate protein and nutrients supports the developing kits.

Preparations for kindling

About a week before kindling, provide a nesting box in the doe cage. The box should be clean, with dimensions of about 30 cm x 30 cm x 30 cm). Line it with straw or hay. The doe may start gathering nesting material, pulling fur from her belly to line the nest, and exhibit restlessness or nesting behaviour. Kindling usually occurs without human intervention. The doe will give birth to a litter of kits, which are born blind, hairless, and helpless. Ensure the kits are warm and that the doe is nursing them. The doe may clean the kits and may eat the placentas.

Post-birth care

Kits rely on their mother's milk for the first few weeks. They should start nibbling on solid food around 2-3 weeks of age. Monitor the kits for signs of health issues, such as not gaining weight or not nursing. The doe usually cares for themselves, ensuring they are kept clean and safe.

Weaning

Kits are usually weaned at 4-6 weeks of age. By this time, they should be eating solid food and can be separated from the doe. Gradually introduce the kits to a diet of hay, fresh vegetables, and pellets to ease the transition from mother's milk to solid food.

Rabbit sexing

Sexing is the process of identifying the sex of the rabbit. Sexing rabbits is essential for managing breeding and ensuring the proper care of rabbits. Table 8.2 shows a guide on differentiating between male and female rabbits.

Table 8.2: A guide on differentiating between male and female rabbits.

Characteristic	Male rabbit (buck)	Female rabbit (doe)
Genital part appearance	Small rounded protrusion (penis) or testicles, if visible	Slit-like vertical opening (vulva)
Testicles	Present in mature bucks; pouch-like	Not present
Behaviour	Territorial, may spray urine, more aggressive	More nurturing, less aggressive
Size and build	May be slightly larger or muscular	May have a rounder, wider build

Activity 8.4

1. Pay a visit to your school farm or a nearby rabbit farms, livestock training and research institutes and perform the following tasks:
 - (a) Inquire on breeding methods and signs of a pregnant doe;
 - (b) Study preparation required before and during kindling; and
 - (c) Observe post-birth care for both doe and kittens.
2. Write a report based on this study.

Exercise 8.3

1. Explain factors affecting breeding in rabbit production.
2. What will happen if a buck is introduced to a doe cage during mating?
3. Describe procedures for sexing in rabbit.
4. Describe the signs of a pregnant doe.

Diseases and parasites control in rabbits

Disease control in rabbits is crucial for maintaining their health and productivity. Effective disease management involves preventive measures, early detection, and prompt treatment.

Preventive measures

Regularly clean and disinfect rabbit cages, pens, and equipment to prevent the spread of diseases. Remove soiled bedding, and wash feeding and watering containers daily. Dispose of waste properly to avoid contamination. Before introducing new rabbits to the existing herd, quarantine them for at least 2-4 weeks. Speak with a veterinarian to determine the appropriate vaccination schedule and vaccines for rabbits. Control fleas, mites, and other external parasites by keeping the environment clean and using appropriate treatments. Regularly check for signs of internal parasites and consult a veterinarian for deworming if necessary.

Symptoms of illness

Some illness symptoms in rabbits include coughing, sneezing, nasal/eye discharge, diarrhoea, bloating, or changes in faecal pellets. Itching, hair loss, or sores. Others are abnormal discharge, difficulty in kindling, and exhibiting poor mothering behaviour.

Treatment and management

If rabbits show signs of illness, consult a veterinarian for proper diagnosis and treatment. Avoid using medications without professional guidance. If a rabbit is ill, isolate it from healthy rabbits to prevent the spread of disease. Ensure sick rabbits have a clean, comfortable space and closely monitor their hydration and nutrition. Table 8.3 shows some common rabbit diseases and their control measures.

Table 8.3: Some common rabbit diseases and their control measures

Disease/ condition	Symptoms	Treatment/ control	Transmission
Myxomatosis	Swelling of eyes, ears, and genitals, lethargy.	There is no cure; prevention is through vaccination.	Direct contact with infected rabbit insects.
Rabbit Hemorrhagic Disease (RHD)	Sudden death, bleeding from nose and mouth.	No specific treatment; vaccination for prevention.	Direct contact with infected rabbits, contaminated environment.

Disease/ condition	Symptoms	Treatment/ control	Transmission
Pasteurellosis (Snuffles)	Nasal discharge, sneezing, respiratory distress.	Antibiotics as prescribed by a vet.	Direct contact with infected rabbit respiratory droplets.
Coccidiosis	Diarrhea, weight loss, poor growth.	Anticoccidial medications; improved sanitation.	Ingestion of oocysts from contaminated feed or bedding.
Ear mites	Scratching ears, head shaking, crusty ears.	Topical mite treatments or medication.	Direct contact with infected rabbits.
Worms (Helminths)	Weight loss, poor coat condition, diarrhoea.	Deworming medications as advised by a vet.	Ingestion of larvae or eggs from contaminated environment.

Activity 8.5

1. Pay a visit to the school or nearby rabbit farm and observe/inquire on the following:
 - (a) Signs of a sick rabbit; and
 - (b) Management measures for a sick rabbit.
2. Write a report.
3. Adopt management measures and apply it in the school rabbitry.

Marketing of rabbit

Rabbit farming can be done for different market needs such as meat production, production of breeding stock, educational and medical laboratory purposes, for pets, show animals, and for wool and skin. The market for the rabbits can be selling live rabbits to commercial processors or laboratory animal suppliers. Alternatively, you may dress and sell the rabbit meat to friends, relatives, neighbours, restaurants, or stores. Each of these markets needs clean, healthy, and well-prepared meat.

Slaughtering of rabbits

Field dressing employed in large animals is applied to rabbits. Slaughtering can be done using any clean, sharp cutting instrument. Alternatively, a hopper popper facility is recommended to dislocate the cervical spine of the rabbit. After slaughter, the rabbit is skinned to get the rabbit meat, as described herein.

(a) Skinning rabbits: The process entails the following steps:

- (i) Make a slot/cut in the skin across the back;
- (ii) Grasp a fold of skin on each side of a cut and pull in opposite directions. The skin pulls away easily; then,
- (iii) Remove the head, feet, and the tail; you have the skinned rabbit with its digestive and respiratory organs.

(b) Dressing of the rabbit: The following procedure demonstrates the process of dressing the rabbit. This procedure applies to all small farm animals like guinea pigs and squirrels. Dressing steps are:

- (i) Place the knife or blade at the anus and cut through the skin and pelvic bone;
- (ii) Cut up through the breastbone while your fingers are placed under the knife to avoid cutting off the internal organs;
- (iii) Separate the esophagus and windpipe together with their entrails from the rest of the body;
- (iv) Clean the cavity. At this point, you have the dressed rabbit meat. Figures 8.4 (a) and 8.4 (b) show rabbit meat and rabbit meat portions, respectively; then,
- (v) Wipe out the cavity and allow it to cool.



Figure 8.4 (a): Dressed rabbit



Figure 8.4 (b): Butchered rabbit

Activity 8.6

1. Visit your school rabbit farm or a nearby rabbit farm, library or internet searches, and inquire/study on:
 - (a) Practices used in rabbit meat preparation; and
 - (b) Market availability for rabbit products (meat and fur) and by-products (skin, manure, and urine)
2. Use Internet search, video clips, and livestock experts to inquire/study on rabbit slaughtering, skinning, and dressing processes.

Exercise 8.4

1. What do you understand by the following terms as used in rabbit production?
 - (a) rabbit meat;
 - (b) hoop hopper;
 - (c) fur; and
 - (d) entrails.
2. Outline factors to consider for producing good quality rabbit meat.
3. Suggest marketing strategies that can be used to increase the market demand for rabbits, its products and by-products.

Record keeping in rabbit production

Record keeping, like in any livestock farming enterprise, is a practical, effective, and efficient way of managing the enterprise. Observations like breeding dates, kindling dates, vaccination and deworming dates, income, and expenses are crucial data that must be documented throughout the year. Record keeping in rabbit farming has many advantages, including the following:

- (a) Expense and income records help to track the profit and cost of running your farm.
- (b) Breeding records tell the farmers if the doe has conceived regularly during the year and if the doe has an excellent mothering ability.
- (c) Medication records indicate the type of diseases, drugs used, and the control measures undertaken.
- (d) Feeding records provide information on the feed types used, amount and costs, and performance, which will help in future feeding plans.

Activity 8.7

1. Visit your school rabbit farm or a nearby rabbit farm and study on:
 - (a) Types of rabbit records kept at that farm; and
 - (b) Usefulness of the records in the rabbit production enterprise.
2. Summarise your observation, in your portfolio.
3. Prepare a plan and carry-out a project in rabbit production from selection of suitable breeds to marketing.



In your portfolio, summarise what you have learnt from this chapter.

Glossary

Artificial insemination (AI)	a breeding technique where semen is collected from a male animal and manually inserted into the female's reproductive tract.
Bacon	a type of meat product that comes from the back or sides of a pig, often cured and smoked.
Bagging	the practice of covering a bunch of bananas with bags to protect the fruits against damage caused by insects and other pests.
Banana bunch	large cluster of bananas growing on a single stalk.
Banana finger	an individual banana within a hand.
Banana hand	a subgroup of bananas on a bunch. Each hand consists of 10-20 bananas.
Barrow	a castrated male pig, usually raised for meat production.
Biosecurity	biological practices designed to prevent the introduction and spread of diseases within an animal unit.
Boar	a mature male pig used for breeding. Boars are selected for their genetics to improve the quality of the pig herd.
Buck	a male rabbit.
Castration	the removal of the testicles of a young male animal to prevent it from becoming a breeding animal (boar).
Circadian rhythms	the natural physical mental and behaviour changes that occur in living organism including sows that follows a 24 hours cycle of which light and dark have biggest influence.

Coccidiosis	a common parasitic disease in rabbits, is caused by protozoa and affects the intestinal tract.
Colony housing	a housing system where multiple rabbits live together in a larger, communal space.
Conception	the process of becoming pregnant, where a sperm fertilises an egg.
Confinement systems	a type of livestock farming where animals are kept in controlled indoor environments. Confinement systems include farrowing crates, nursery units, and finishing barns. An intensive pig production system is an example of a confinement system.
Conformation	the physical shape and structure of an animal e.g. rabbit is often evaluated in shows or breeding programs.
Creep feeding	the practice of providing supplemental feed to young animals while they are still nursing. This helps them transition to solid food and promotes faster growth.
Crop rotation	the practice of growing crops in alternation to improve soil health, optimise nutrient and combat pests and weed pressure.
Crossbreeding	the practice of breeding animals e.g. pigs/ rabbits from two different breeds to combine desirable traits from both. This often results in hybrid vigour, where the offspring perform better than their parents.
Culling	The process of removing animals e.g. rabbits or pigs from the breeding stock due to poor health, productivity, or undesirable traits.
Curing	a post-harvest process that involves storing harvested crop produce in warm, humid

Cuttings

conditions to heal any damage to the skin and improve flavour and shelf life.

the sections of stems that are planted to propagate new plants.

Dam

adult female animal e.g. rabbit for breeding purpose

Dessert banana

bananas that are sweet in flavour eaten fresh and raw, after the main part of the meal.

Doe

a female rabbit.

Estrus (Heat)

the period when a female animal e.g. a pig is fertile and ready to be bred. Sows and gilts usually come into heat every 21 days.

Farrowing

the process of giving birth in pigs. This is a critical time for both the sow and her piglets, requiring careful management.

Farrowing index

a measure of the reproductive performance of a sow, often calculated as the number of litters produced per year.

Feed Conversion Ratio (FCR)

a measure of an animal's efficiency in converting feed into body mass., This is important in evaluating the productivity of an animal.

Feeder pig

a young pig that has been weaned and is being fed to gain weight before it is sold for finishing or slaughter.

Finishing

the stage in pig production where pigs are fed to reach market weight, typically around 90-100 kg. This is the last phase before slaughter.

Flushing

the practice of increasing the feed provided to female animals e.g. sows and gilts before breeding to improve their body condition and increase the number of eggs released during ovulation.

Forage	plant materials, such as hay, grass, or leafy greens, that animals like rabbits eat.
Fostering	the practice of transferring piglets from one sow to another to ensure that all piglets receive adequate milk and care. This is often done when a sow has too many piglets to nurse.
Fur block	a condition where rabbits ingest too much fur, leading to a blockage in their digestive system.
Fur breeds	Breeds that are primarily raised for their fur, such as the Angora rabbits
Gestation	the period between conception and birth in animals; in pigs, typically lasting about 114 days (3 months, 3 weeks, 3 days). It is around 28-31 days for rabbits
Gilt	a young female pig that has not yet given birth. Gilts are typically bred when they reach maturity.
Grooming	the process of cleaning and maintaining a rabbit's fur, which can include brushing, trimming, and bathing.
Harvesting	the process of collecting mature crops from the field.
Herbivore	an animal that primarily eats plants, as rabbits do.
Herd	a group of animals e.g. pigs raised together.
Hutch	a type of cage or enclosure where rabbits are housed.
Inbreeding	breeding between closely related animals e.g. rabbits or pigs, often to maintain specific traits within a line, but it can also lead to health issues.

Kindling

the process of a doe giving birth to her litter.

Kit

a baby rabbit, usually referred to as a kit until it is weaned.

Lactation

the period during which a female animal e.g. a sow produces milk to feed her piglets. This usually lasts until the piglets are weaned.

Laxative feed

a type of feed that helps improve bowel movement and feed intake hence, minimises incidences of constipation.

Litter of pigs

a group of piglets born to a sow at one time. The size of a litter can vary but typically ranges from 8-12 piglets. Modern prolific pig breeds can produce many piglets up to above 20 piglets per sow per farrow.

Litter of rabbit

the group of baby rabbits born to a doe at one time.

Market weight

the ideal weight at which an animal is considered ready for slaughter, usually between 90-100 kg in pigs. Reaching market weight efficiently is a key goal in pig production.

Mating ratio

the ratio of male to female animals during breeding. This is important to ensure that all female animals are bred within a specific timeframe.

Meat breeds

animal breeds that are specifically raised for their meat production.

Molting

the process by which rabbits shed their fur, usually occurring seasonally.

Monoculture

the agricultural practice of growing a single species of crop over a large area for many successive growing seasons.

Nest box	a box provided to the doe where she can give birth and care for her kits.
Offspring	the young of an animal.
Parasites	organisms like mites, fleas, or worms that can infest animals like rabbits and pigs, causing health problems.
Parturition	the act of giving birth. It has the same meaning as farrowing.
Pedigree	a recorded lineage or ancestry of an animal, often used for breeding purposes.
Pellet feed	type of concentrated feed made from ground-up grains and other ingredients, formulated to meet the nutritional needs of animals e.g., rabbits.
Piglet	young pig, usually under the age of 3-4 weeks, before it is weaned from its mother.
Plantain	the type of banana that is starchy rather than sweet and is often used in cooking
Porcine	a term relating to pigs or swine. It is often used in veterinary medicine and scientific research.
Prophylactic	a treatment designed to prevent disease occurrence.
Pseudostem	the false stem or trunk of the banana plant.
Quarantine	the isolation of new or sick animals to prevent the spread of disease to the rest of the herd.
Rabbitry	a facility or farm where rabbits are bred and raised.
Run	an outdoor space where rabbits can exercise and graze.
Setts	sections of cane stalks with buds used for propagation in sugarcane.
Sexing	the process of determining the sex of a rabbit, usually performed on young rabbits.

Sire	adult male animal e.g., rabbit for breeding purposes.
Sow	a mature female pig that has given birth to piglets. Sows are the cornerstone of pig production as they produce the offspring/piglets.
Supplemental feeding	the practice of providing additional feed, such as grains or vegetables, alongside the primary diet of forage.
Sword suckers	a banana sucker that has narrow, sword-shaped leaves, more vigorous.
Tail docking	the practice of cutting off a portion of a piglet's tail to prevent tail biting, a common issue in pig herds that can lead to injury and infection.
Mat	the entire banana plant system, including the mother plant, corm, and all suckers growing around it.
Trait	a certain characteristic that can produce a particular type of behavioural in an animal
Tuber	an enlarged storage root which is rich in starch and serves as the primary edible part.
Vaccination	the process of administering vaccines to an animal to help its immune system develop immunity against a disease.
Vines	sweet potato planting materials.
Weaning in pigs	the process of gradually removing piglets from depending on the sow's milk to solid feed. This usually occurs at about 3-4 weeks of age.
Weaning in rabbit	the process of transitioning kits from their mother's milk to solid food.
Wire floor	a type of flooring in rabbit cages that allows waste to fall through, keeping the living area clean.

Bibliography

FAO (n.d). Post Harvest Management of Banana for Quality & Safety. Extension Brochure.

Fielding, D. (1991). *Rabbits. The Tropical Agriculturist*, CTA MacMillan.

Gamborg, C., and Sandoe, P. (2005). Sustainability in farm animal breeding. *A review. Livestock Production Science. 92, 221-231.*

Harvest Plus (n.d). *Sweet Potato Seed Systems*. A Farmer Trainers' Guide.

Helen Keller International Tanzania ((2012). Orange-Fleshed Sweet Potato Situation Analysis and Needs Assessment Tanzania Report (2012). <http://www.sweetpotatoknowledge.org/wp-content/uploads/2016/02/Tanzania-OFSP-Situation-Analysis-Report.pdf>

Hillocks, J.R, Thresh M. J. and Bellotti C A. (Eds) (2002). “*Cassava: Biology, Production and Utilization*,” CABI Publishing.

Ikani, I.E., and Dafwang, I.I. (1995). *Pig Production Technology for Piggery Farmers*. Extension Bulletin No. 25 Livestock Series No.1. National Agricultural Extension and Research Liaison Services, Ahmadu Bello University.

JICA (2016). Smallholder Horticulture Empowerment & Promotion Project for Local and Up-Scaling (SHEP PLUS). *Banana production*. SHEP PLUS.

JICA (2019). Smallholder Horticulture Empowerment & Promotion Project for Local and Up-Scaling (SHEP PLUS). *Potato production*. SHEP PLUS.

Kapinga, E.R, Ewell, T.P and Kileo R. (1995). *Sweet potato in Tanzania and Food Systems: Implications for research*. International Potato Centre.

Khosa, E., and Villane, P. S., (2012). *Modern Agriculture Learner's Book Form 3*. Oxford University Press Southern Africa (Pty) Ltd.

- Kroschel, J., Mujica, N., Okonya, J., & Alyokhin, A. (2020). *Insect pests affecting potatoes in tropical, subtropical, and temperate regions*. In H. Campos & O. Ortiz (Eds.), *The Potato Crop: Its agricultural, nutritional and social contribution to humankind* (pp. 251–306). Springer. <https://doi.org/10.1007/978-3-030-28683->
- MAFS (2003). *Poultry and rabbit husbandry. Technical handbook No.7*. The United Republic of Tanzania.
- Ndunguru, J., Kapinga, R., Sseruwagi, P., Sayi, B., Mwanga, R., Tumwegamire, S., & Rugutu, C. (2009). Assessing the sweet potato virus disease and its associated vectors in northwestern Tanzania and central Uganda. *African Journal of Agricultural Research*, 4(4), 334–343.
- Okorie J.U., (1983). *Agricultural Science for Teacher Training Colleges*. Longman Group UK Limited.
- Shilunga, S., Kennedy, L. Jomanga, E.K. (2021). The Status of Banana Production in Tanzania; a Review of Threats and Opportunities. *International Journal of Current Science Research and Review* 4:1260-1275.
- Sibuga K.K, Lekule F.P, Kinala T.Z (2004). *Principles and practices of livestock production*, Educational Books Publishers LTD.
- Tanzania Institute of Education (2021). *Agriculture for Secondary Schools, students Book Form One* Tanzania Institute of Education.
- Tanzania Institute of Education (2021). *Agriculture for Secondary Schools. Students Book Form Two* Tanzania Institute of Education.
- Tanzania Institute of Education (2022). *Agriculture for Secondary Schools, Students Book Form Three*. Tanzania Institute of Education.
- TOSCI (2020). *Potato catalogue of Tanzania*. TOSCI